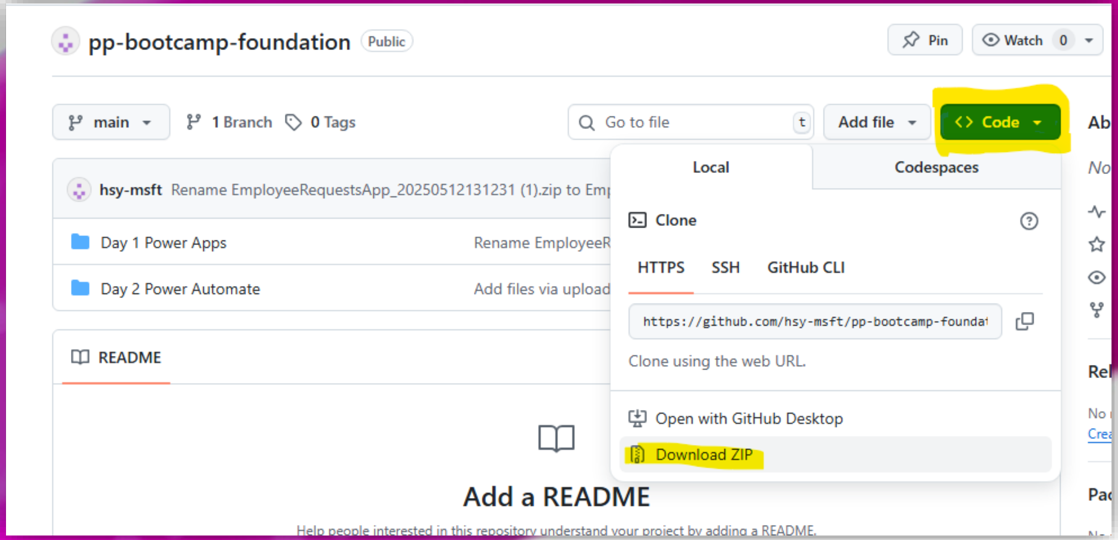
Scan the QR code below or enter URL below to take your attendance

https://forms.office.com/r/8Cn9CpMEYV

If you have any use cases you'd like support with, feel free to reach out to your respective Digitalisation Office rep! ☺

Download Lab Materials

https://github.com/KwangEng/PP-Bootcamp-Material



Microsoft Power Platform

Power Platform Bootcamp

Sing Kwang Eng
Cloud Solution Architect – Business Applications
Microsoft

Agenda

Time	Activity Title
0930	Kick Off
0940	Power Platform Overview
1000	Power Apps Overview
1030	Break
1045	Power Apps Screens, Controls and Formulas
1130	Lunch
1300	Hands on Exercises
1500	Break
1515	Hands on Exercises
1700	End

Session Information

Target Audience

New to Power Platform and creating applications. Basic knowledge of Microsoft 365 applications such as Microsoft PowerPoint, Microsoft Teams & Microsoft Excel.

Objectives

- Gain a foundational understanding of Microsoft Power Platform, including the core components
  - Power Apps
  - Power Automate
  - Power BI
- Learn how to create simple, low-code applications, automate processes, and analyse data using built-in tools.
- Attempt hands-on exercises and explore how these tools can be used to solve real-world business challenges and drive productivity, while gaining confidence in building scalable solutions without extensive programming knowledge.

Power Platform Overview

Microsoft Power Platform

Power Platform is a rapid application development platform that helps citizen developers with limited coding abilities, to innovate and build new solutions faster in an environment and user experience that feels like PowerPoint and Excel, called a low-code environment.

Power Apps	Power Automate	Copilot Studio	Power Pages	Power BI
Application development	Process automation	Intelligent chatbots	Website	Business analytics

Data connectors

AI Builder

Microsoft Dataverse

Power Fx

Why Low Code Development?

750 million new apps need to be built in by 2025[1]

75% of apps to be built on Low Code[4]

Power Platform

33 million monthly active users[2]

25% reduction in time to complete tasks[3]

Used by **97%** of Fortune 500

Traditional Software is slow and expensive

Pursuing Digital Transformation

Security & Compliance Risks

1. IDC, Connected Development: Delivering Digital-First Excellence. October, 2022.
2. Gartner, Forecast Analysis: Low Code development Technologies. January, 2021.
3. Microsoft Fiscal Year 2023 Third Quarter Earnings Conference Call. April, 2023.
4. Gartner, Forecast Analysis: Low-Code Development Technologies. October 2022

A Platform For All Makers

Citizen Developer, IT Administrator, Professional Developer, App Wizard. We’ve got you

Intuitive to use & easy to learn for Citizen Developers

Build apps fast with a point-and-click approach to app design. Choose from a large selection of templates or start from a blank canvas

Easily connect your app to data and use Excel-like expressions to easily add logic

One admin center to rule them all for IT Administrators

No compromises on governance and security. One centralized view and management of all your 1st & 3rd party apps, eliminating shadow IT

GDPR compliance, and enterprise grade security, consistent with the experience across O365 & D365

World class Pro Developer support and ALM

Experience the full range of development and ALM functionality with rich pro developer tools like Visual Studio and DevOps

Reuse your current IP and skills – Power Apps component framework, CLI and VS Code create instant value for the business

Power Apps Overview

What are Power Apps?

- Applications that anyone can build for the web and mobile devices—
  No coding experience.

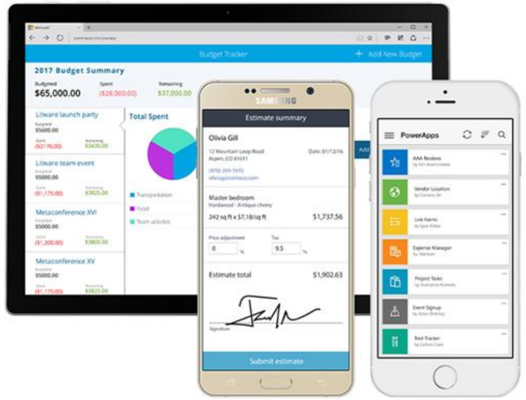
- A **drag and drop** interface allows you to add and configure what you need to solve the problem at hand with controls like buttons and galleries.

- An **Excel-like formula** bar allows you to dig in a little deeper and apply more complex rules and logic. See results in real time.

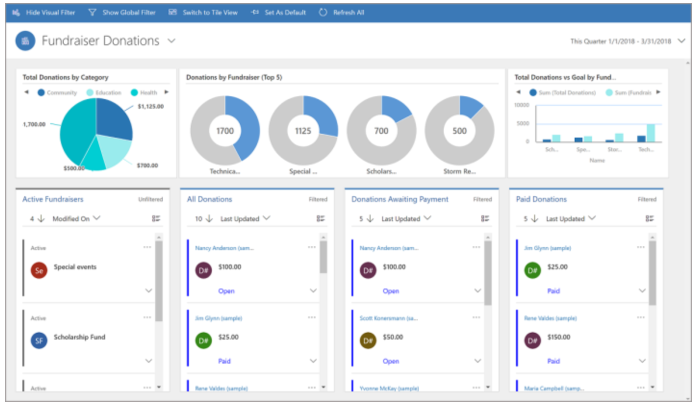
- You can **extend Power** Apps by creating customized connections and building blocks that can be used across multiple programs

- Use the App Via Microsoft Teams, on your phone or the on web

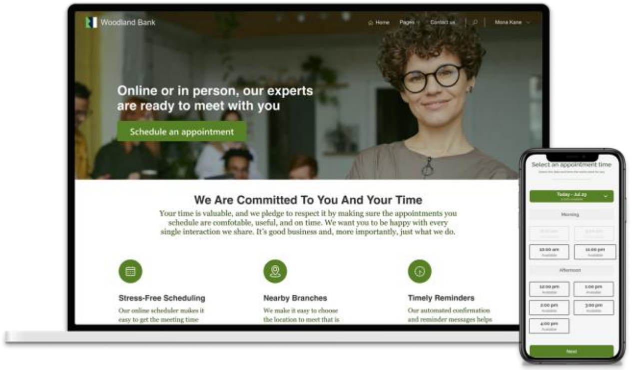
Types of Power Apps



Canvas apps start with user experience, with pixel-perfect control to build highly customized task- and role-based apps that mash up data from 700+ sources



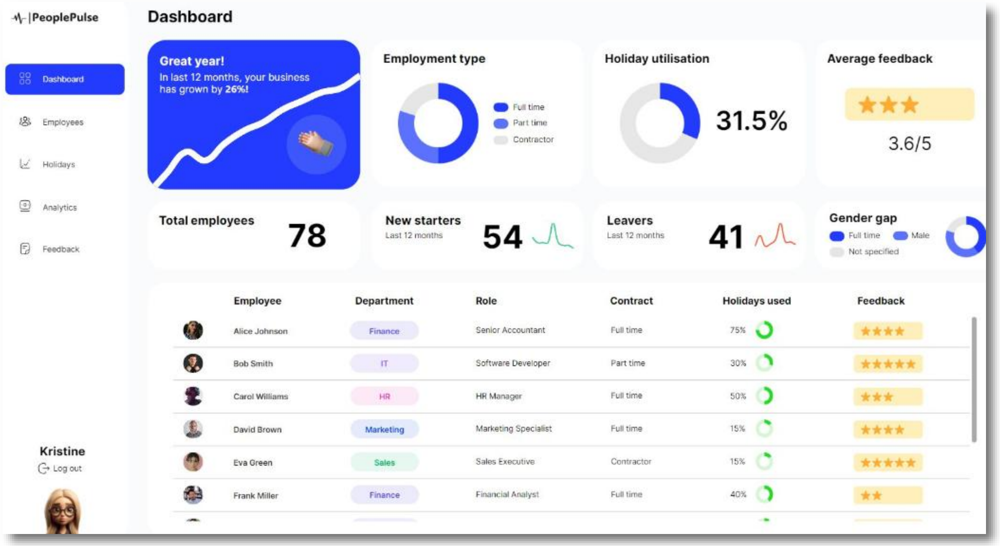
Model-driven apps leverage your data model, relationships, and business processes to automatically generate immersive, responsive applications



Power Pages an easier way to create modern, secure, responsive business websites to rapidly engage your customers, partners and communities

The focus should be on Canvas Apps today as the others tend to more complex. It is sufficient to understand the use case for the others.

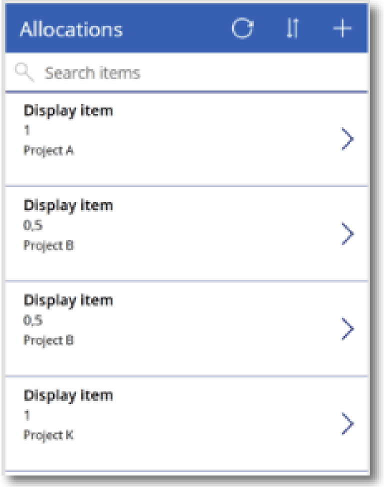
Power Apps Examples

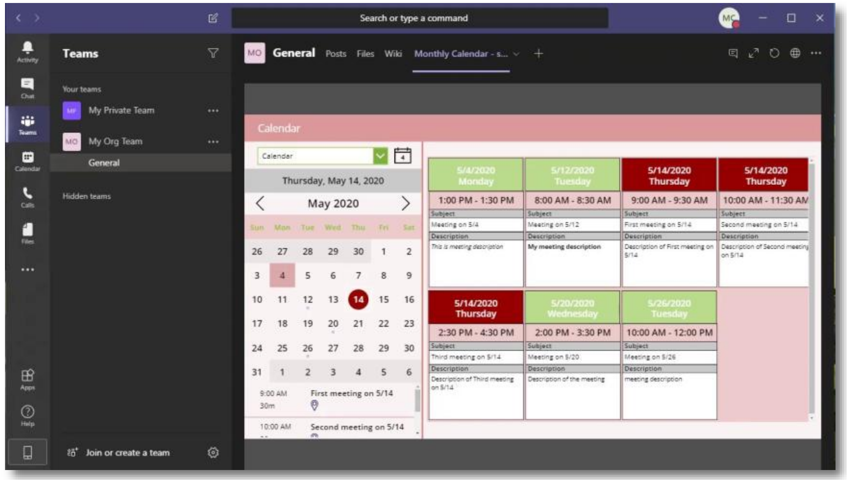


Build modern applications that can aid business decision making

Use apps as an entry point to drive automation or visualize information

Create simple mobile ready applications

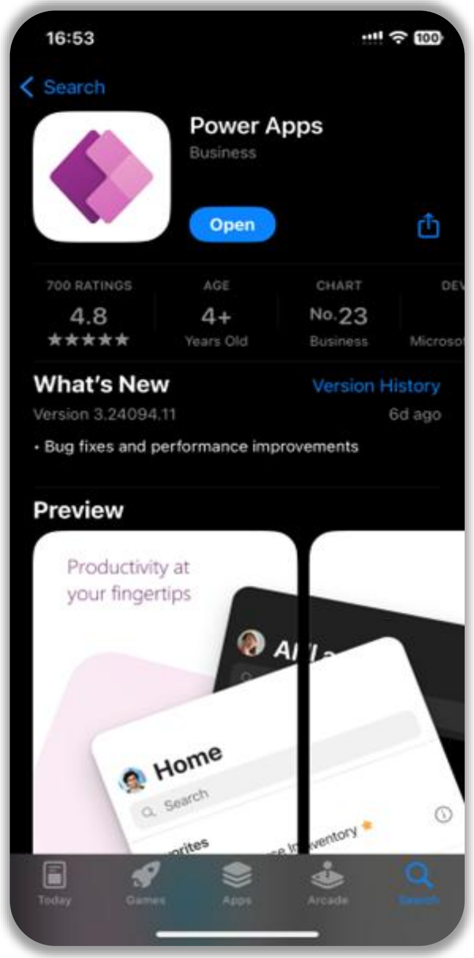


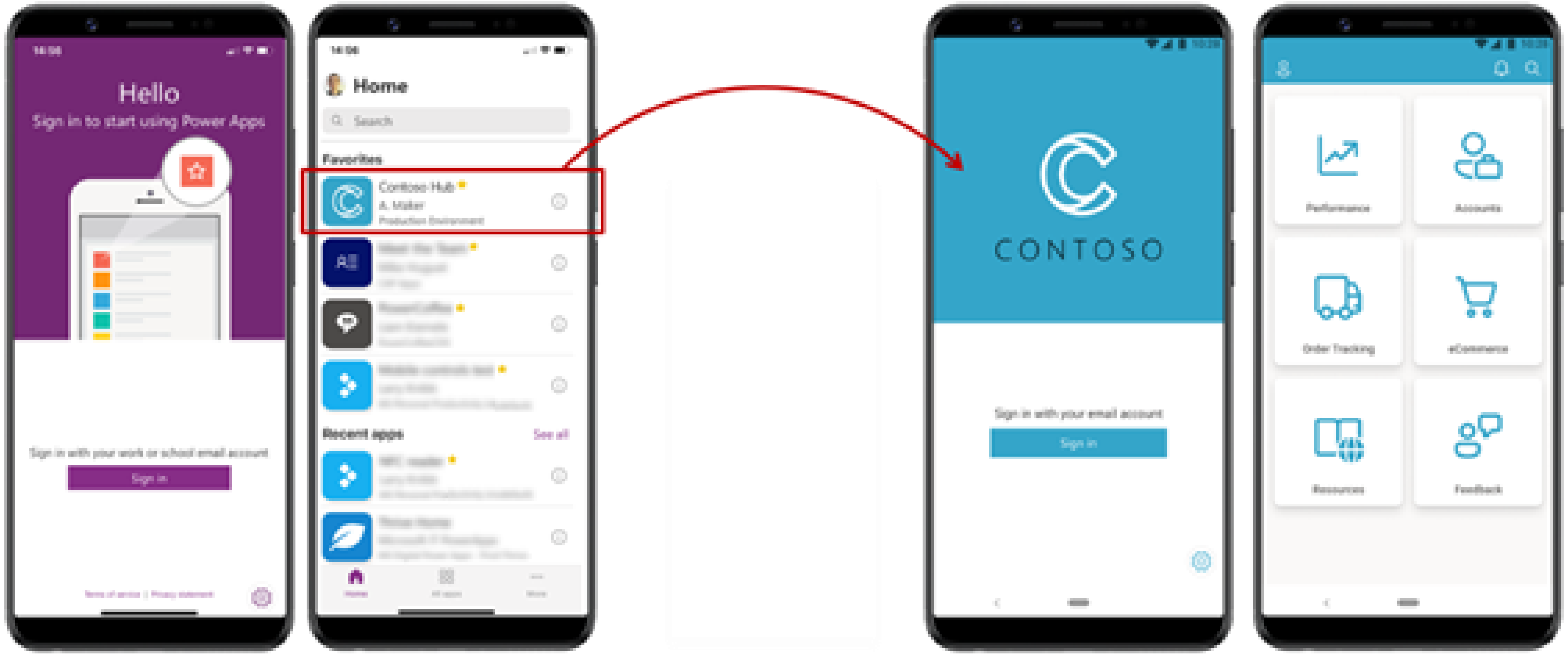


Access apps from a wide variety of sources

Power App mobile access

Access multiple Power Apps via the “Power Apps” App



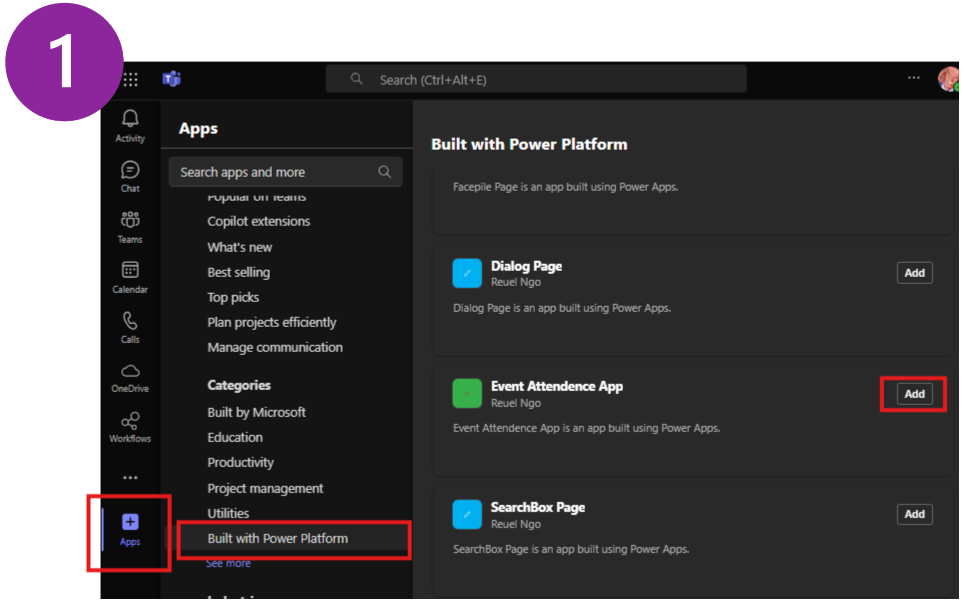


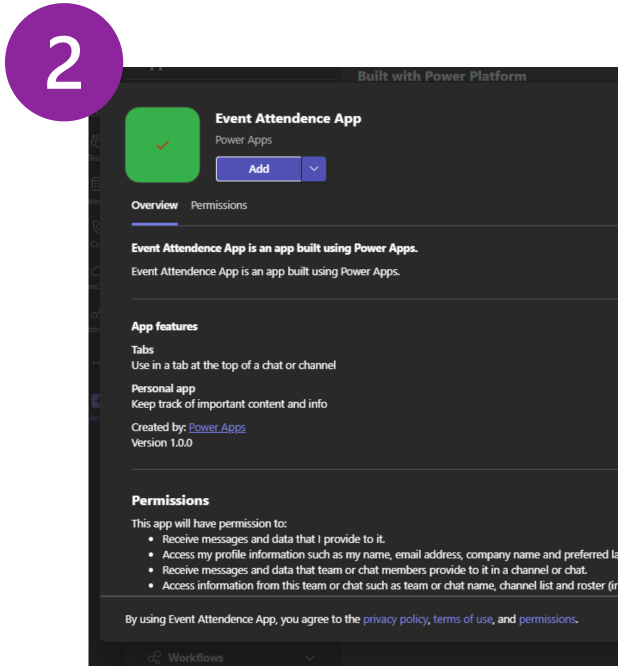
Accessing Power Apps in Teams – Pinned on Sidebar

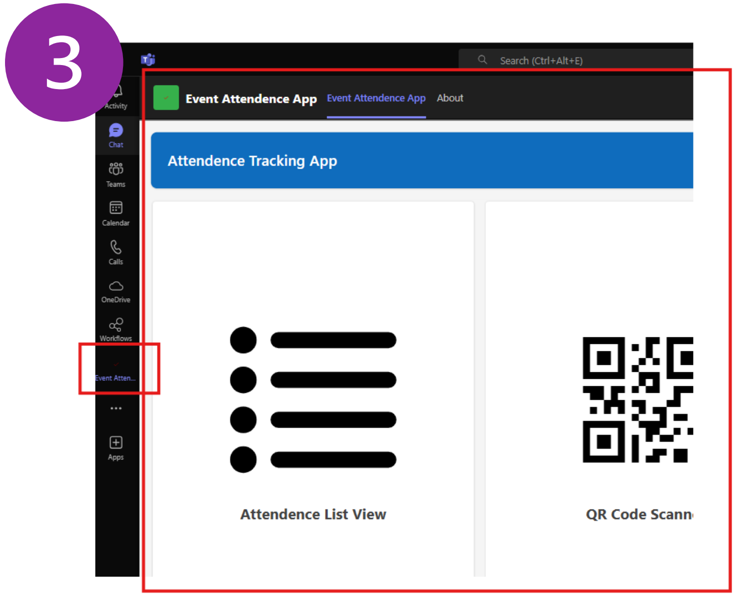
Add a Power App to the side bar for easy access in Microsoft teams

1. Search for the Published App in Teams
2. Click on Add
3. Access it directly from Microsoft Teams.

With this, users can quickly access essential apps without leaving Teams. This is good for frequently used apps.







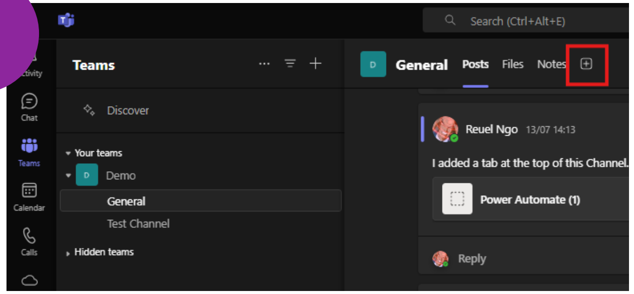
Accessing Power Apps in Teams – Pinned on a Channel

Add a Power App to a channel for easy group access in Microsoft teams

1. Navigate to your channel of choice and click on add.
2. Click on Power Apps
3. Search for the App you wish to add
4. Use the App

When Apps are critical for team collaboration, pinning an app in a channel allows users to use relevant apps without leaving the channel.

1



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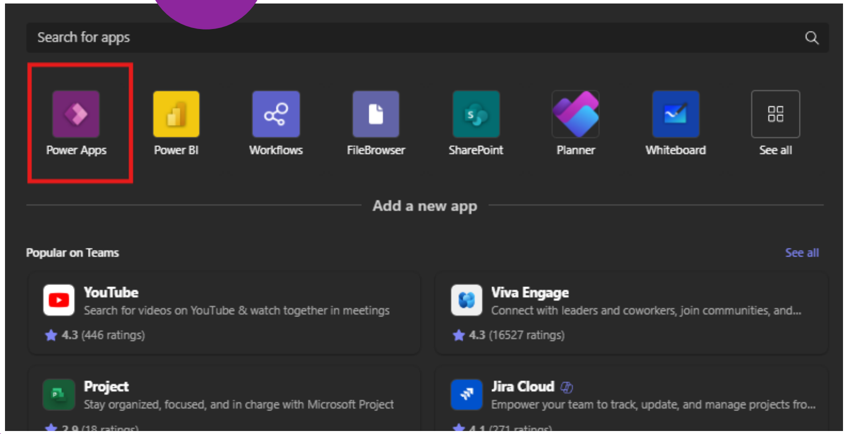

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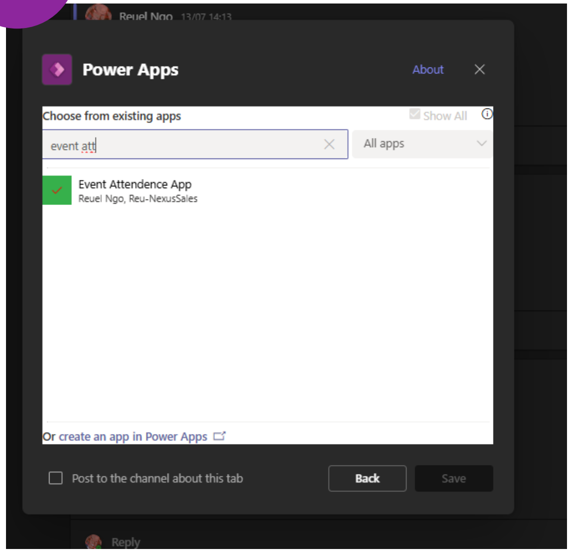

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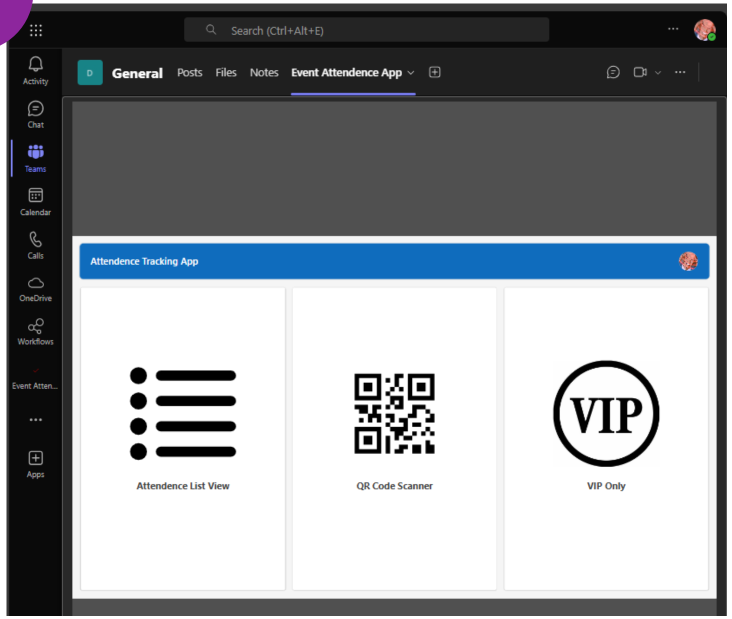


Power Apps

Connectors

Types of connectors

- Prebuilt Connectors – ready-made connectors that allows you to connect to apps and data in the cloud
  - Standard Connectors – Requires a Power Apps Seeded License
  - Premium Connectors – Requires a Power Apps Premium License
- Custom Connectors – Connectors that you build to support your apps or data that might not be residing in the cloud

Popular Standard connectors

- Excel Online (OneDrive)
- Office 365 Outlook
- OneDrive for Business
- Microsoft Teams
- Microsoft To-Do (Business)
- Office 365 Users
- Microsoft Forms
- Power Apps Notification V2
- SharePoint
- Planner

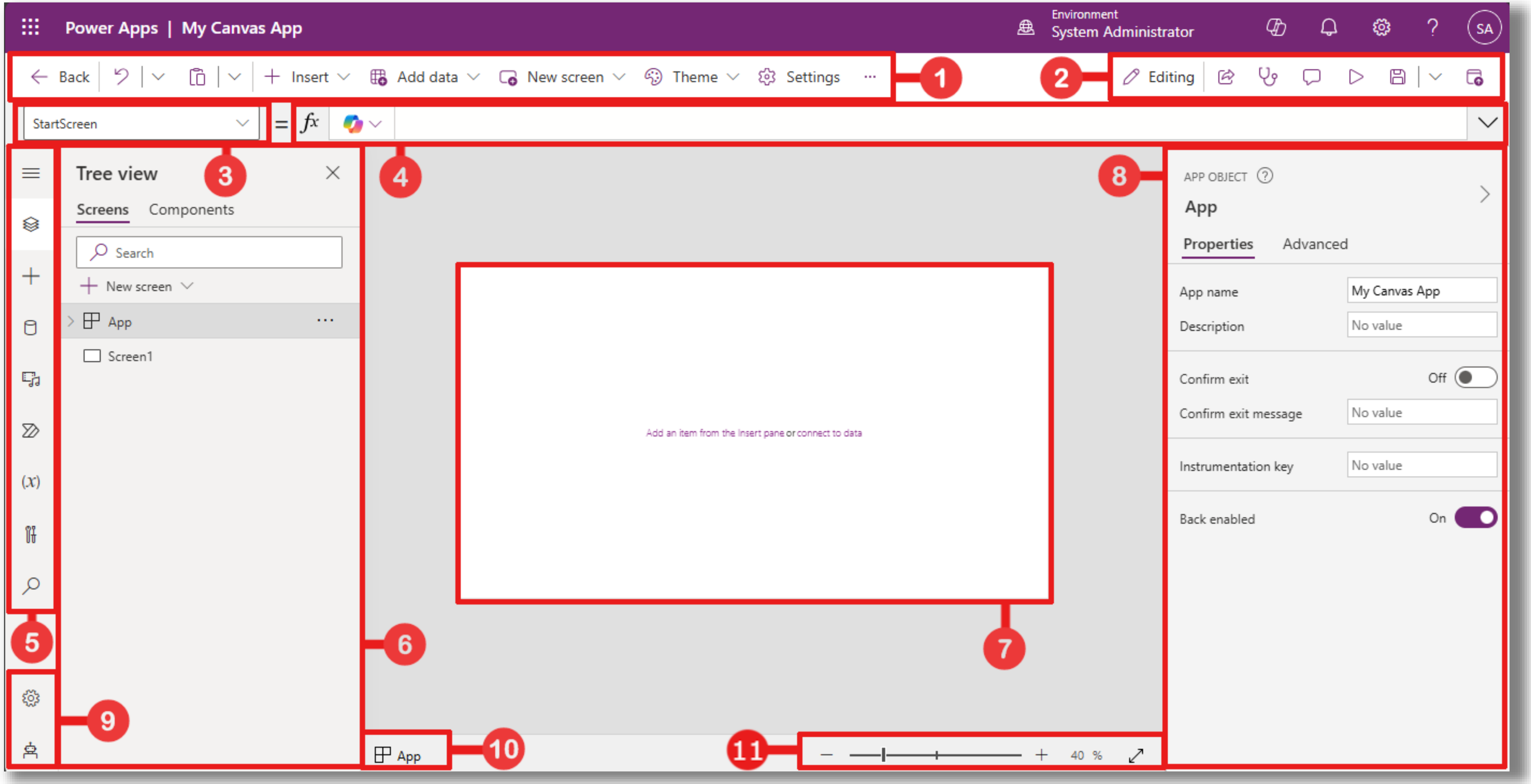
[List of supported connectors | Microsoft Power Automate]

Power Apps

Screens and Controls

Power Apps Studio: The Designer's Toolbox

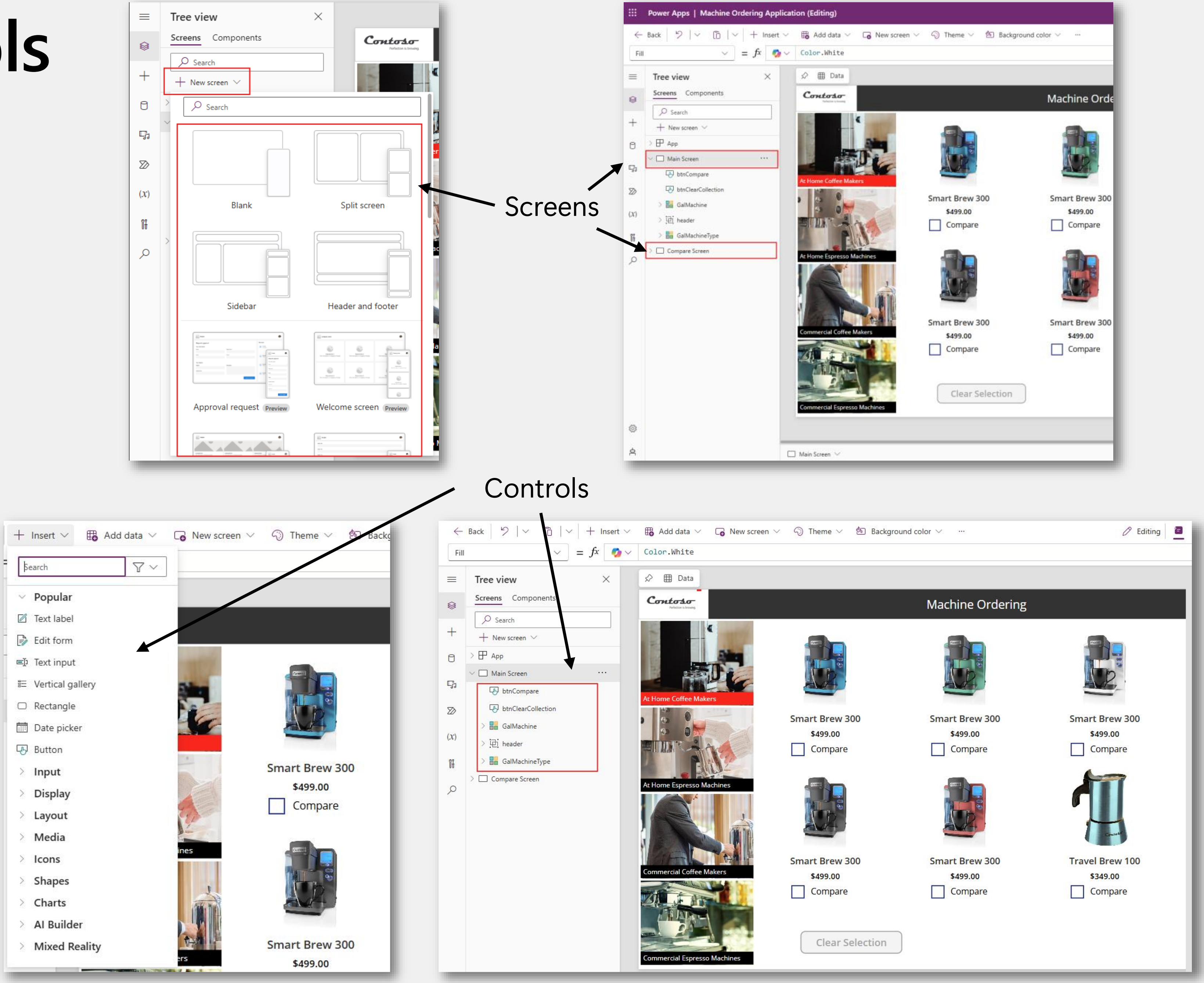
You can use Power Apps Studio to design, build, and manage your canvas app.



1. Power Apps Studio modern command bar
2. App actions
3. Properties List
4. Formula bar
5. App authoring menu
6. App authoring options
7. Canvas/screen
8. Properties pane
9. Settings and virtual agent
10. Screen selector
11. Change canvas screen size

Screens and controls

- Screens – Most apps have multiple Screen controls that contain Label controls, Button controls, and other controls that show data and support navigation.
- Controls – UI elements that you can control the properties of in your app.



Screens and controls – Continued

- Screens – Most apps have multiple Screen controls that contain Label controls, Button controls, and other controls that show data and support navigation.
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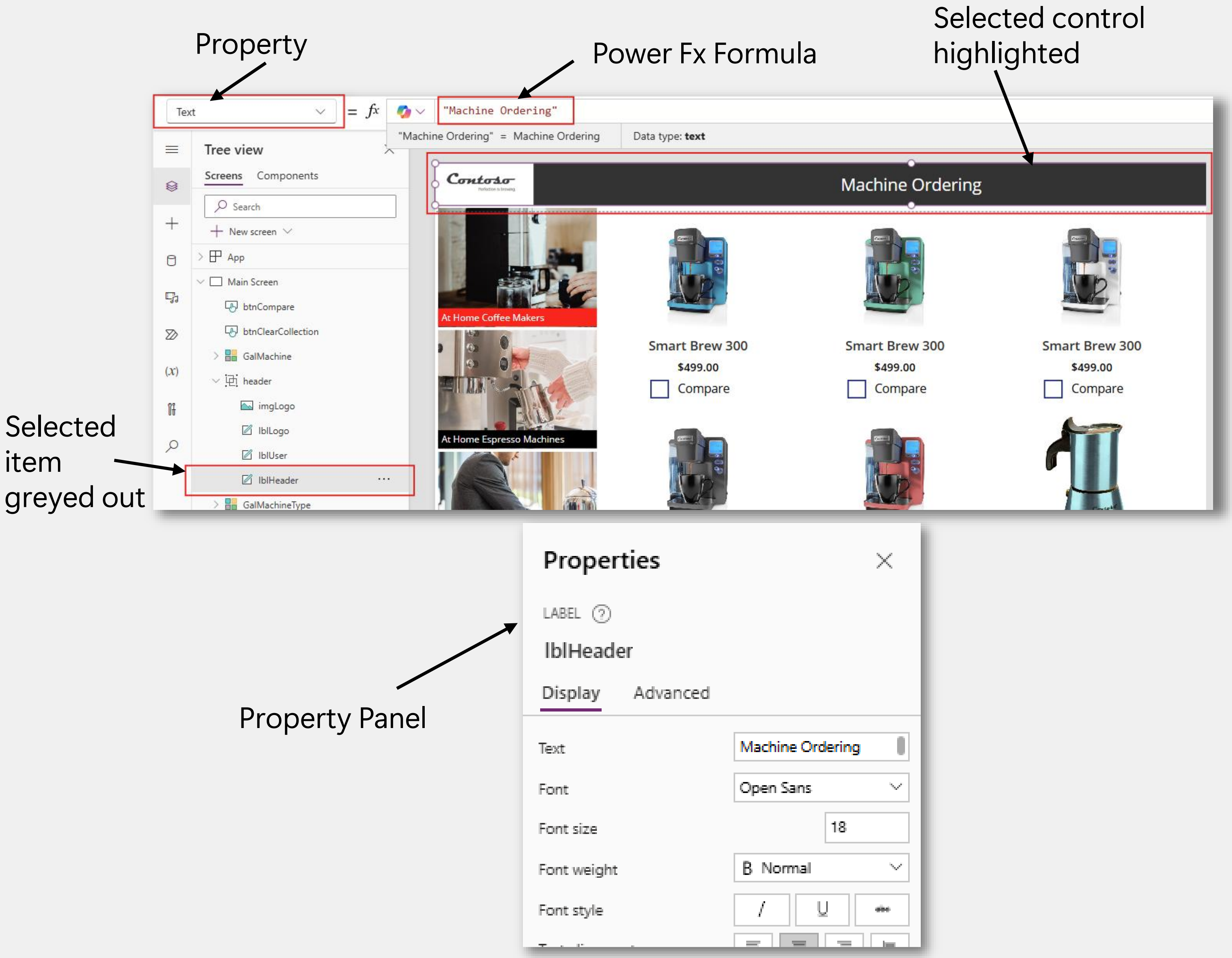
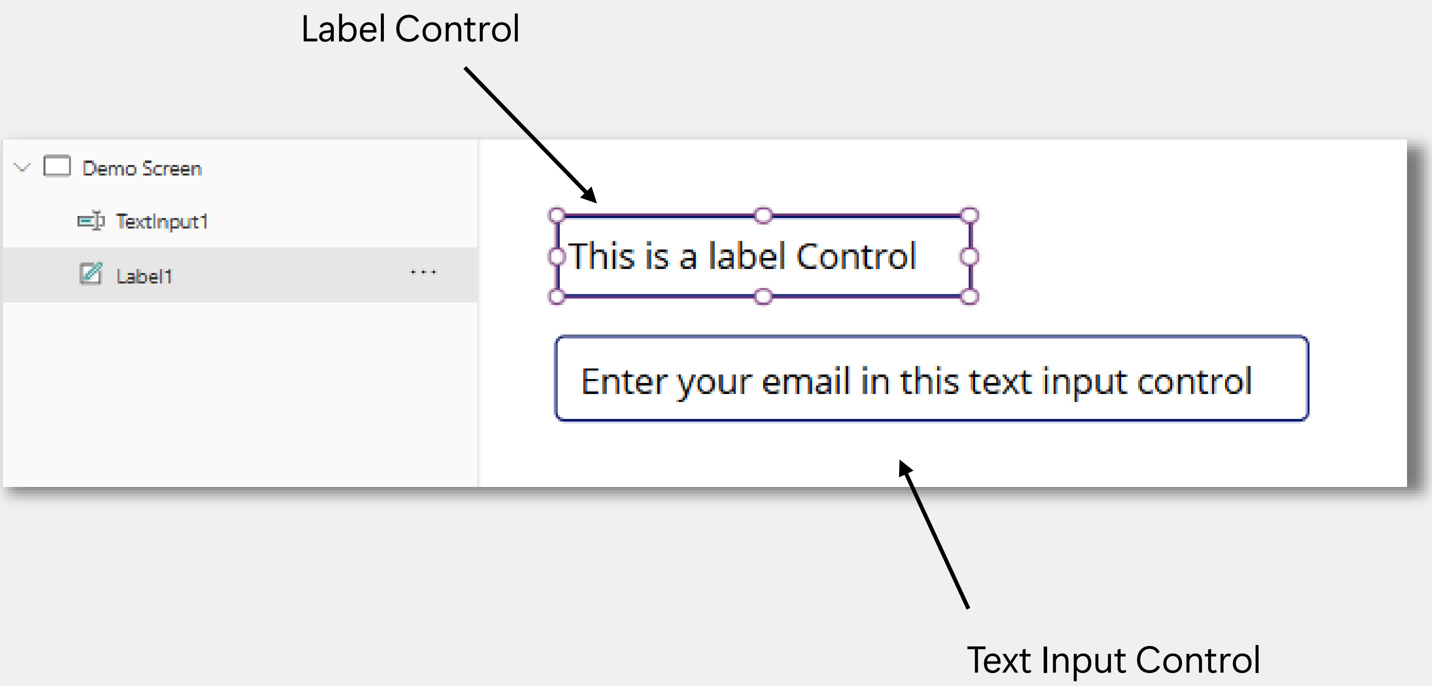


Text input and Label Controls

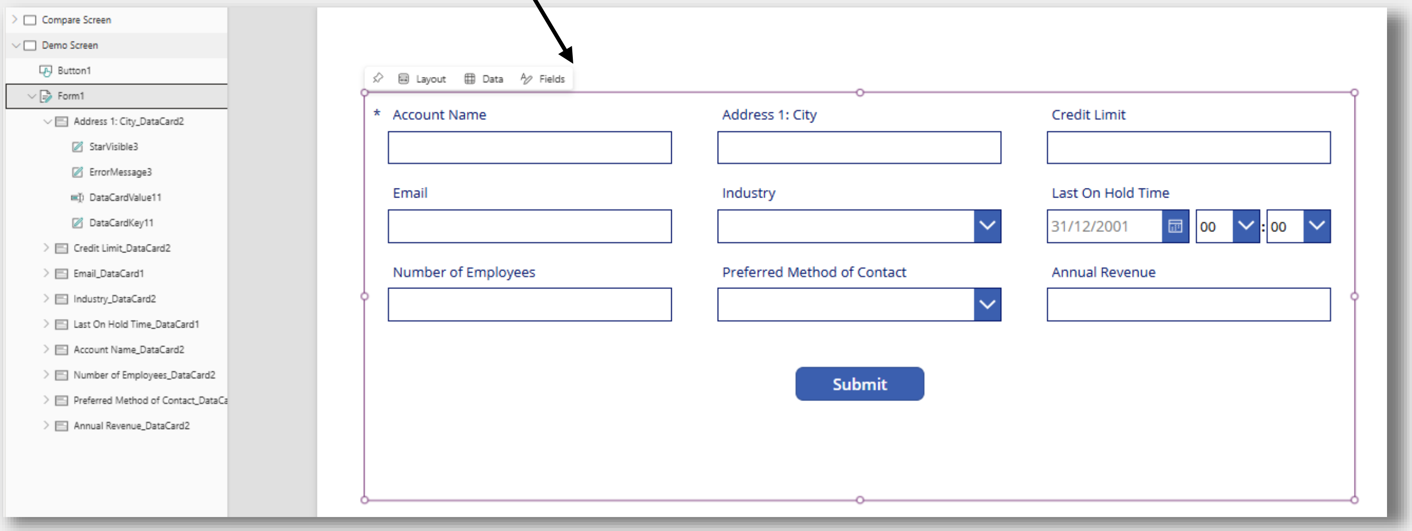
- Label Control – Shows data such as text, numbers, dates, or currency.
- Text input Control – A box in which the user can type text, numbers, and other data.

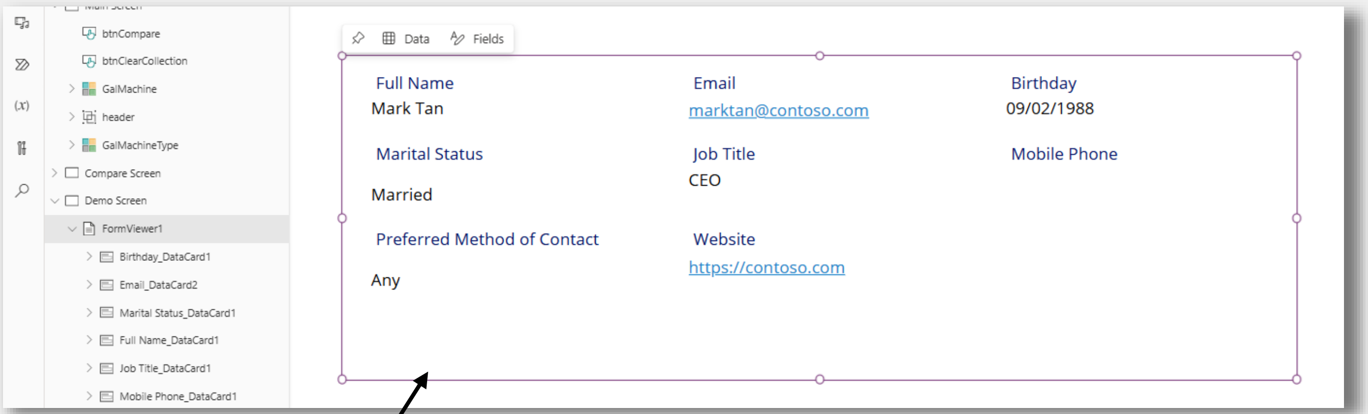


Forms

- Display Form Control – A user can display all fields of a record or only the fields that you specify
- Edit Form Control – A user can edit those fields, create a record, and save those changes to a data source

Edit Form Control

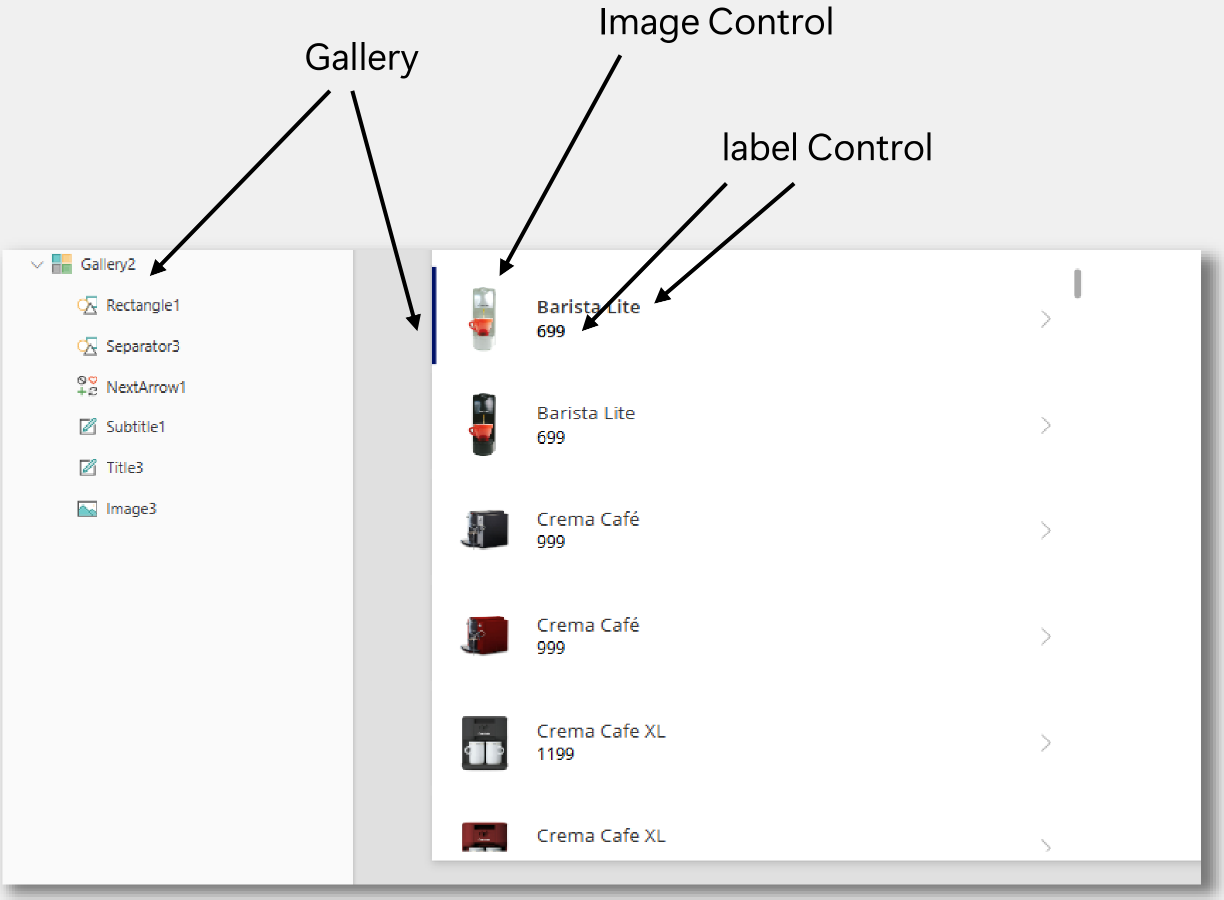




Display Form Control

Galleries

- A Gallery control can show multiple records from a data source, and each record can contain multiple types of data.
- Each data field appears in a separate control which you can configure within the Gallery control.
- The template appears as the first item inside the gallery:
  - On the left edge of a Gallery control in horizontal/landscape orientation.
  - And at the top of a Gallery control in vertical/portrait orientation.



Power Apps

Formulas

What are functions in Power FX?

A function is like giving instructions to an app in a specific order to get a certain result. Just like how you would follow steps when doing something in real life, a function expects you to:

- Give the right inputs: These are the things you need to get the result you want (like the options in an order).
- Follow a specific format: You have to provide the inputs in the correct sequence so the app knows what you're asking for.
- Receive an expected outcome: Based on the inputs you provide, the function will give you a result.

For example, if you order a bubble tea, you can't just say random words like "sugar pearls ice tea," the cashier would be confused. You need to give instructions in the right order:

- What flavour do you want?
- How much sugar?
- How much ice?
- What toppings?

If we were to model ordering bubble tea like a function, it would look something like this:

OrderBubbleTea(flavor, sugarLevel, iceLevel, toppings)

- **flavor** is first because that’s the main part of your order.
- **sugarLevel** and **iceLevel** are next, because they modify how your drink is made.
- **Toppings** come last, and they are **optional additions**.

Function Examples

```
OrderBubbleTea( "classic milk tea", "50%", "regular ice", "pearls" )
OrderBubbleTea( "green tea", "25%", "less ice")
```

Function Executed

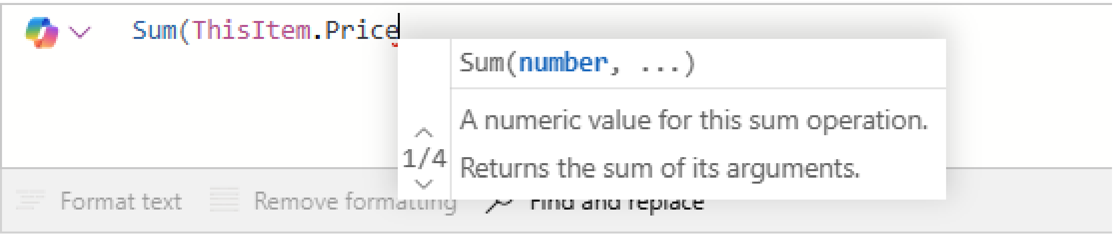
```
OrderBubbleTea( "milk tea", "50%", "half ice", 2 )
OrderBubbleTea( "pizza", "100%", "less ice", "pearls" )
OrderBubbleTea( "taro", "50%", "no ice", “extra ice", "pudding")
OrderBubbleTea( "50%", "half ice")
OrderBubbleTea( “Mala", “Friday", "less ice", "pearls" )
```

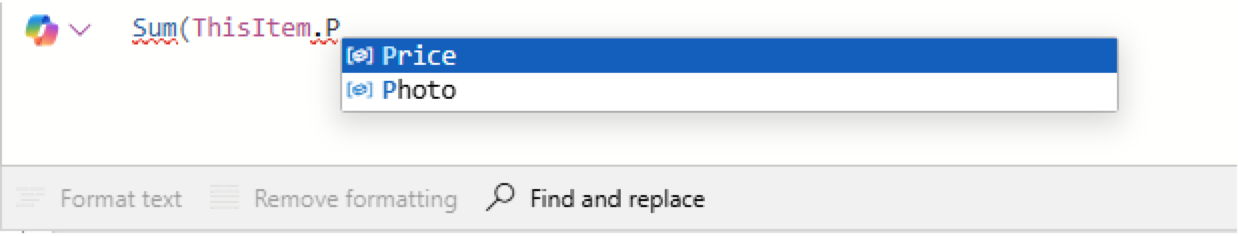
Function Error

PowerApps formulas – think Excel

And	Clock	DateDiff	Location	Notify	Set
App	Collect	Error	LookUp	Now	Sort
As	Color	Exit	Lower	Patch	Sum
Average	ColorValue	Filter	Max	Rand	Text
Back	Concatenate	Host	Mid	RandBetween	Today
Blank	Concurrent	If	Min	Refresh	User
Boolean	CountIf	IsMatch	Mod	Reset	Value
Calendar	CountRows	Launch	Navigate	Self	Year

Suggestions and Completion help





[Formula reference - canvas apps - Power Platform | Microsoft Learn]

Types of Power Fx Formulas

Functions

- Take parameters, perform an operation, return a value
- E.g. Text(ThisItem.Date, “mm-dd-yyyy”)

Signals

- No need to take parameters. Will give an output.
- E.g. UTCNow()

Enumerations

- Returns a pre-defined list of constant values (choices)
- E.g. Color is a enumeration that had pre-defined values such as Color.Blue, Color.Black, etc.

Named Operators

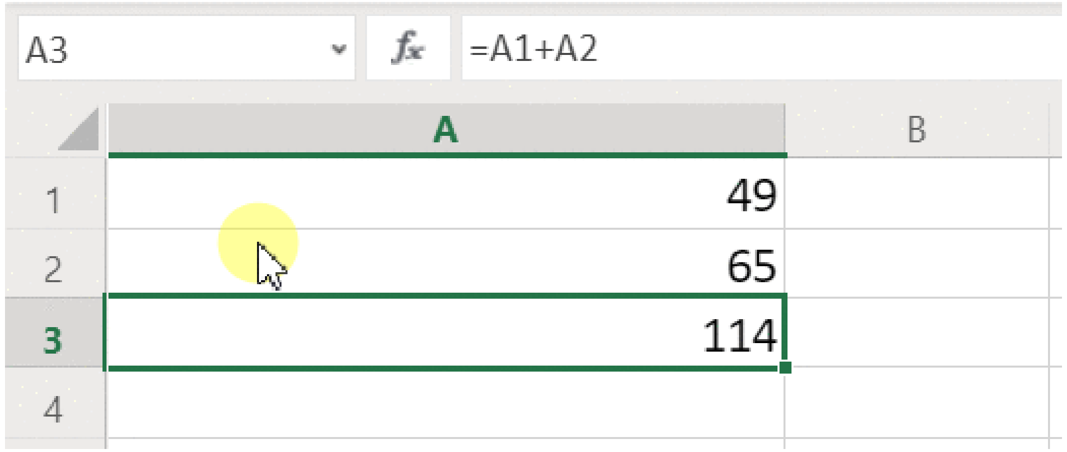
- Reserved keywords that are used by PowerApps
- E.g. ThisItem / Parent

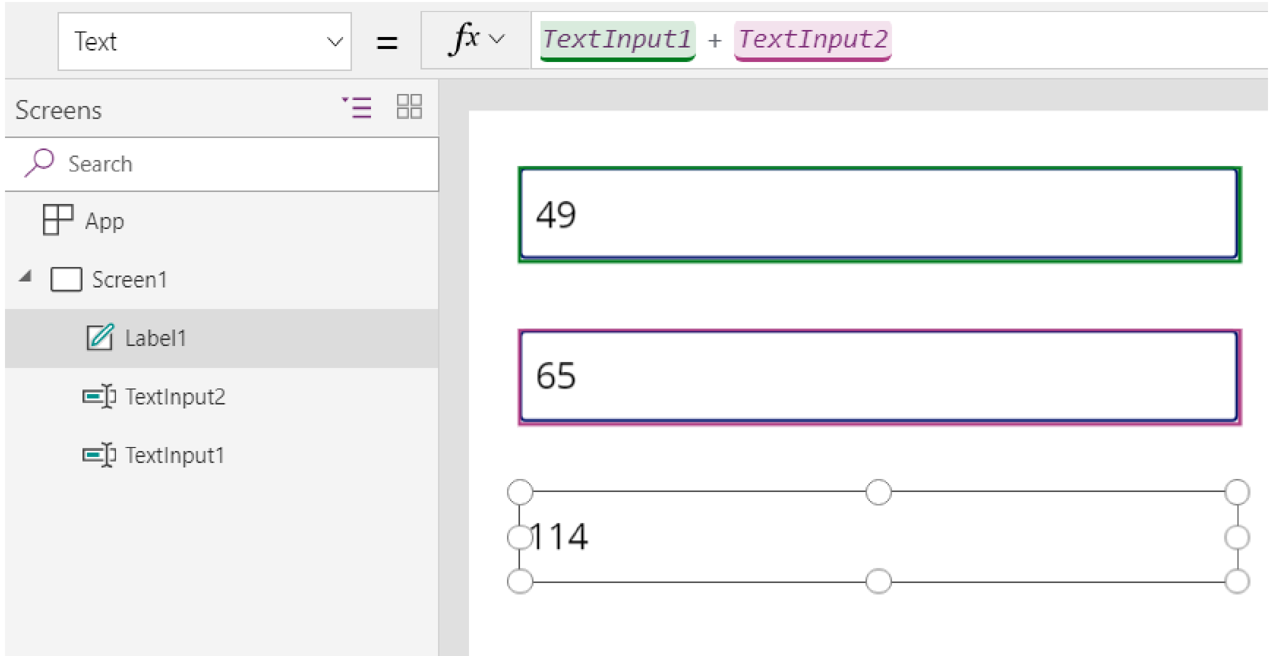
Power Apps

Variables

Variables

- Variables temporarily hold data that can change dynamically
- Types include name, date/time, email address, etc.
- Like Excel formulas





Data types

A data type is an attribute associated with a piece of data that tells a computer system how to interpret its value

- Number – 123.456
- Text – “Hello World”
- Boolean – True or False
- DateTime – 1/12/2017 12:00 PM
- Date – 1/12/2017
- All can hold blank (null) values

Why we need data types

Data types are like categories that help computers (and us) understand what kind of information we're working with. Just like in real life, where different items belong to specific categories, computers need to know what kind of data they're dealing with so they can handle it properly.

1 + 1 = 2

"1" + "1" = 11

With the wrong data type, a **Text Concatenate operation** is performed instead of a **mathematical addition operation**

Item Name	Quantity Sold
Chicken Rice	Five
Nasi Lemak	Six

"Five" + "Six" = "FiveSix"

Without numerical data types, instead of adding the quantity sold, a **Text Concatenate operation** is performed instead of a **mathematical summation**

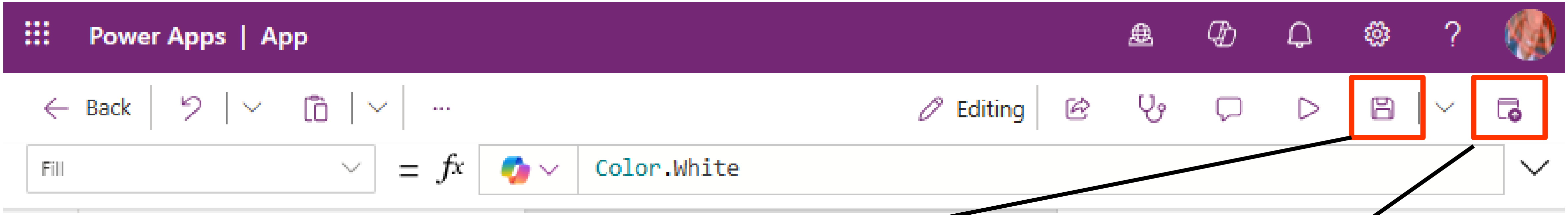
Data types should be the **SAME** for Operations

"Five" + 6 = *Error*

Power Apps

Publishing an App

Saving vs Publishing



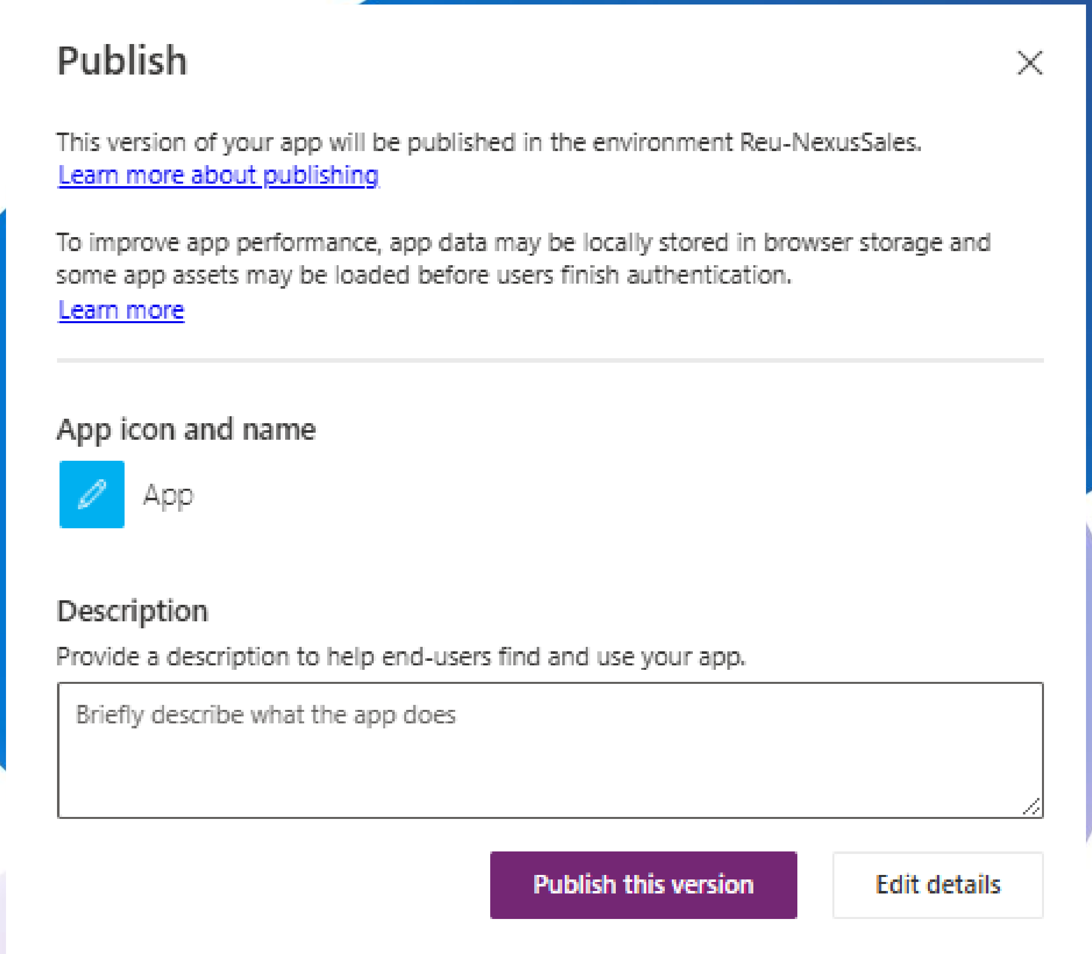
Saving – saves the new changes made to the app and increase the version number. This makes it available to all makers as the latest version.

Publishing – Releases the new version to all users so that they can see the new changes and use new features. E.g. New screens / buttons.

*** Save frequently, Publish only when you have fully tested an app and are confident the formulas and functionality works!

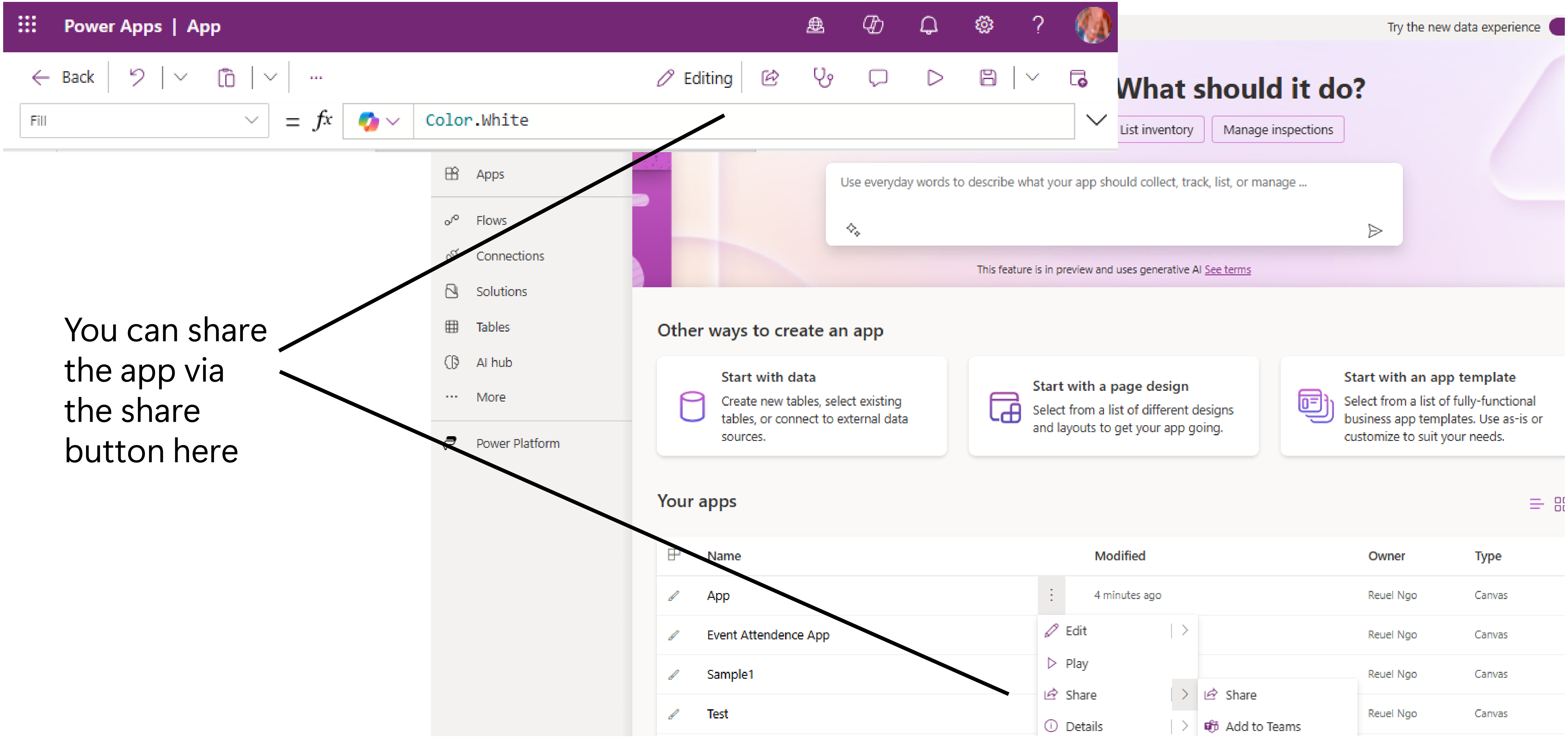
Publishing the App

- You may choose to change the following here:
- Name
- App icon
- Description



Sharing the App

You can share the app via the share button here



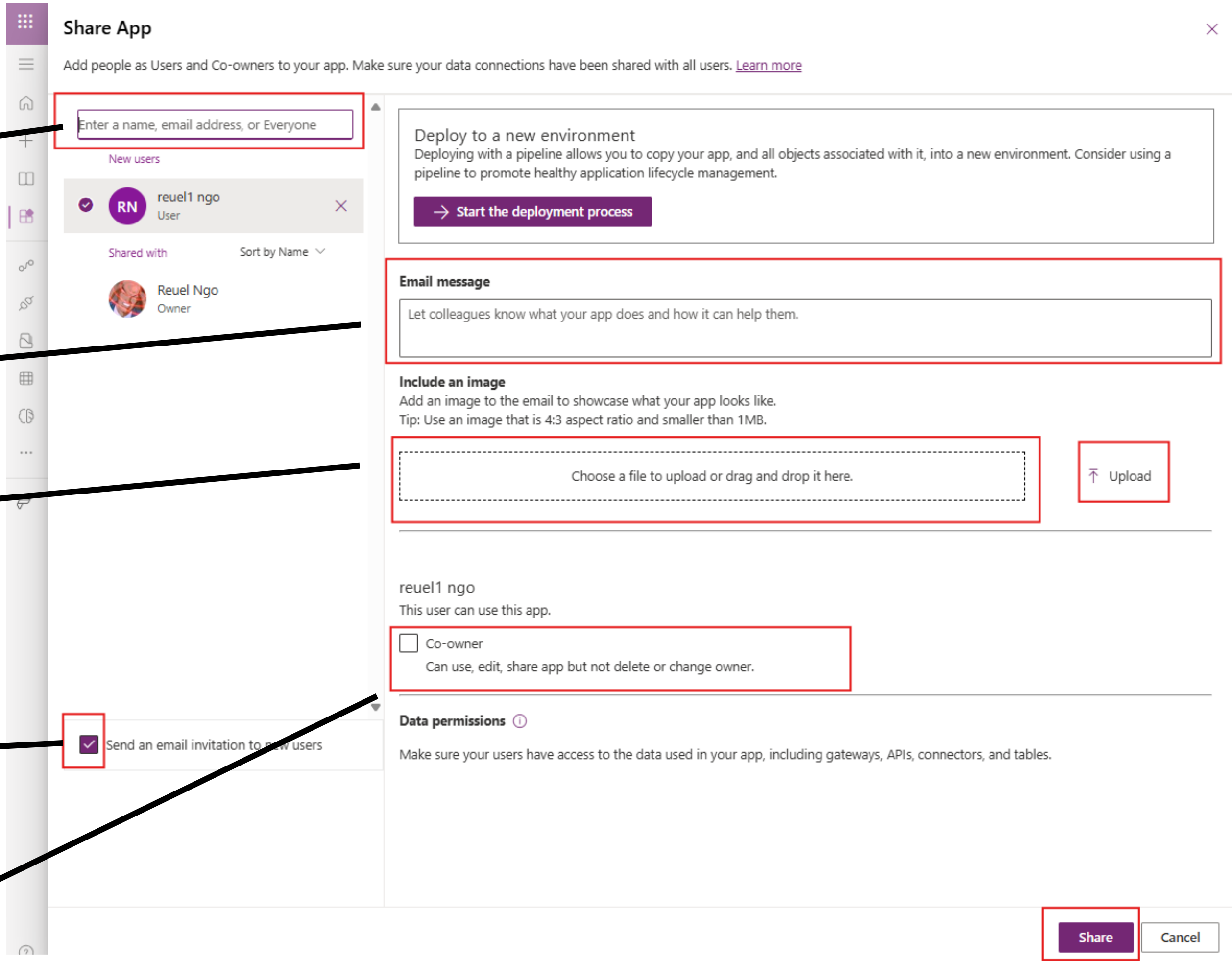
Sharing the App

Specify whom you want to share the app with by name

Send them an email

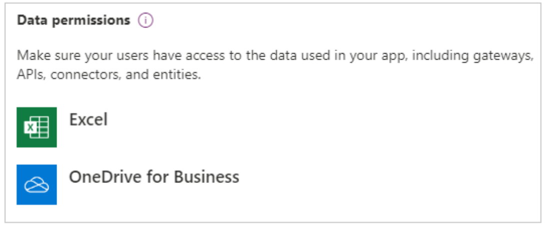
Include an image in the email

Include email invitation checkbox



If you want the user to edit and share the app, select the co-owner checkbox
Best practice*

If you use certain data sources in your app, make sure your users have access to the data sources as well



Microsoft Power Platform

Power Automate Fundamentals

Sing Kwang Eng
Cloud Solution Architect – Business Applications
Microsoft

What is Power Automate?

Time	Activity Title
0900hrs	Introduction to Power Automate
0930hrs	Exploring Power Automate Designer Studio, Copilot and common use cases
1015hrs	Tea Break
1100hrs	Hands on Power Automate Flow Labs – Part 1
1115hrs	**Lunch**
1245 hrs	Project Time

What is Power automate?

A comprehensive, end-to-end cloud automation platform powered by low code and AI that automates actions across the most common enterprise apps and services.

Basic Automation examples

- Post a message to an owner whenever a task is assigned to them in Microsoft planner
- Automatically save all email attachments from Outlook to OneDrive
- Generate and send weekly reports with SharePoint lists, excel
- Convert documents into pdf files when they are dropped into a folder
- Send emails from an excel list with user emails
- Getting approvals from manager

Advanced Automation examples

- Generate insights via GPT from scheduled email briefings
- Identify and notify users who create bad data in SharePoint lists
- Contract renewal reminders

Overview of Power Automate

Power Automate
Product

Flows Automate your workflows	Enhancements Enhance your flows	Apps Create your flows
Cloud flows Digital process automation (DPA) *Automate cloud apps, services, and data with APIs*	**AI Builder** Artificial intelligence (AI) *Add intelligence to your cloud flows*	**Power Automate for web** Web *Create/manage all automation from a central portal.*
Desktop flows Robotic process automation (RPA) *Automate your desktop's user interface (UI)*	**Process advisor** Process mining, task mining *Identify opportunities to automate*	**Power Automate for desktop** Windows *Create and manage desktop flows in Windows*
Business process flows Business process automation (BPA) *Automate multistep, guided business processes*		**Power Automate for mobile** iOS, Android *Create cloud flows on the go for iOS and Android*
		Power Automate for Teams Microsoft Teams *Create cloud flows in the hub for teamwork*

Cloud Flows – Digital Process Automation

Industry Term

Digital process automation (DPA)

Description

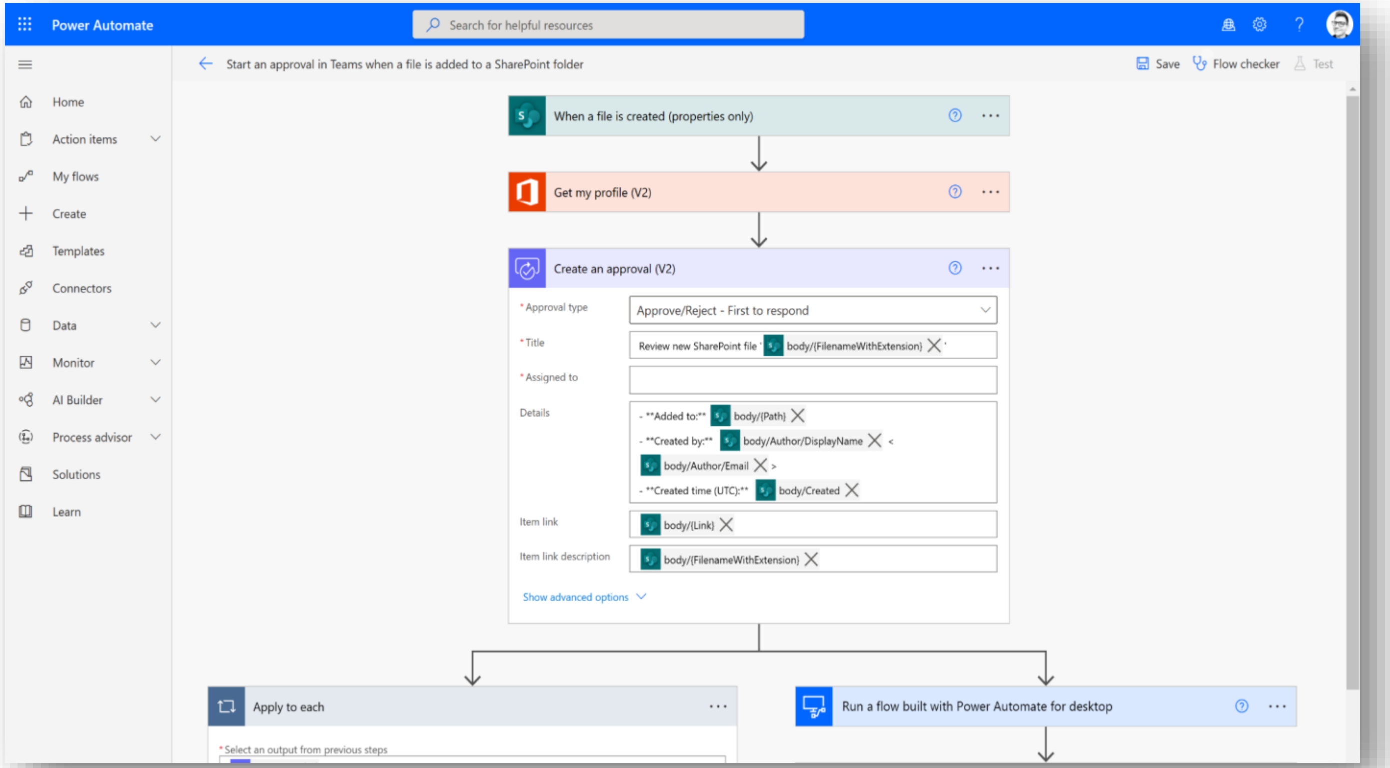
Automate between cloud-based apps, services, and data using API connectors and hundreds of prebuilt templates

Apps

Power Automate for web

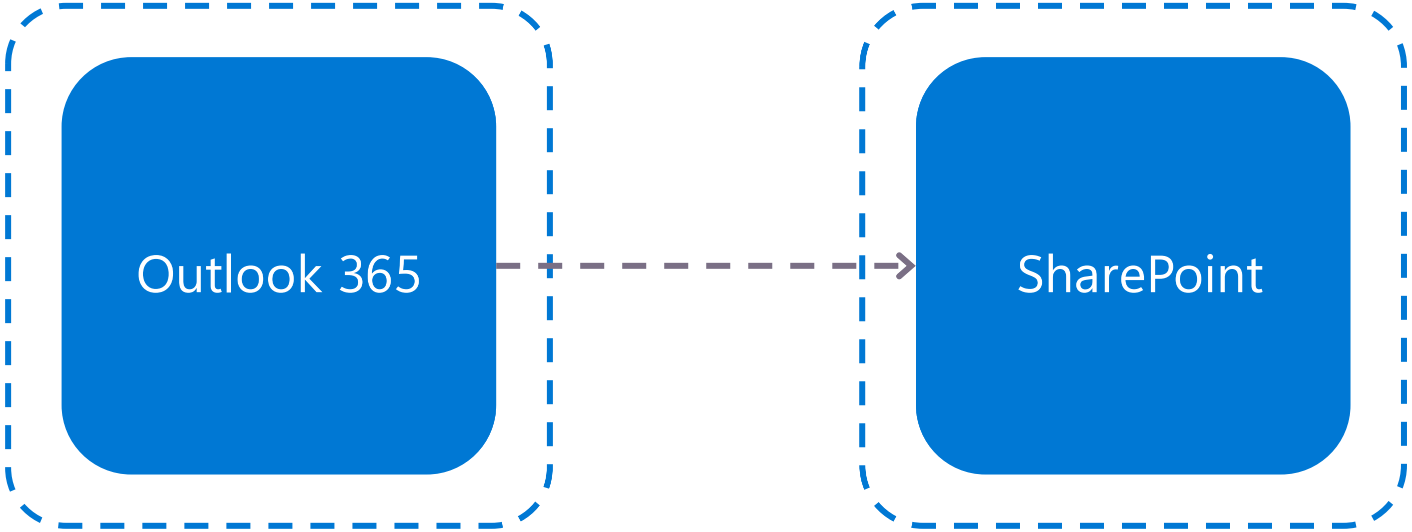
Power Automate for mobile

Power Automate for Microsoft Teams



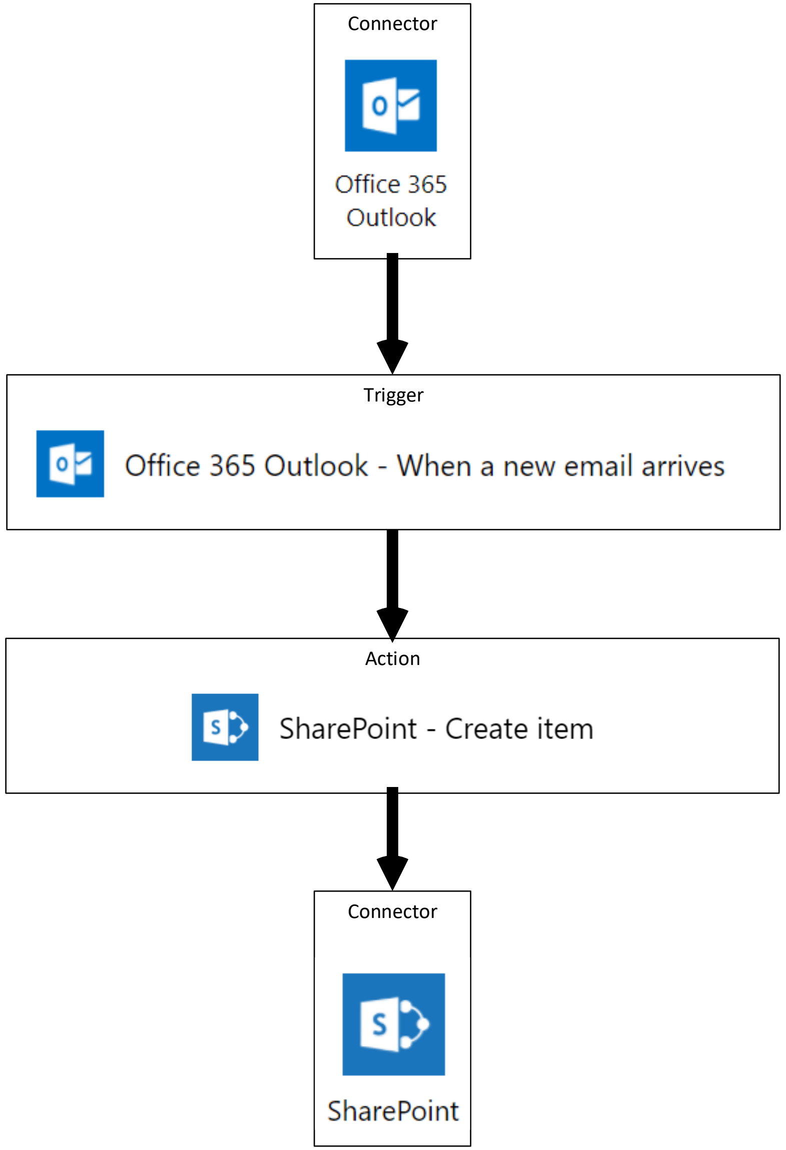
How cloud flows work

Automate API cloud apps, services, and data using connectors



Trigger
Office 365 Connector
Application programming interface (API)

Action(1..n)
SharePoint Connector
Application programming interface (API)



A closer look at cloud flows

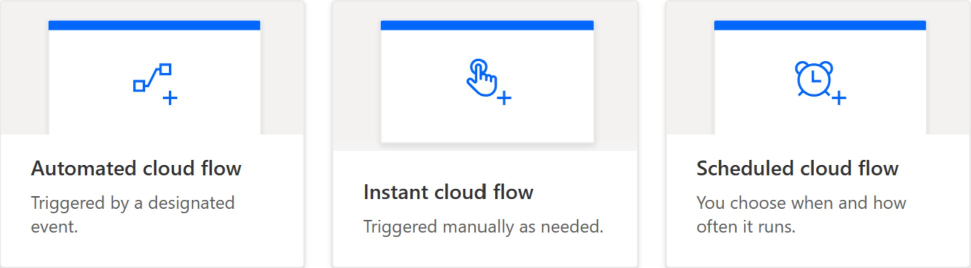
Triggers

Choose how your cloud flows run

Automated
Run a cloud flow when something happens

Instant
Run a cloud flow manually

Scheduled
Run a cloud flow on a schedule



Connectors

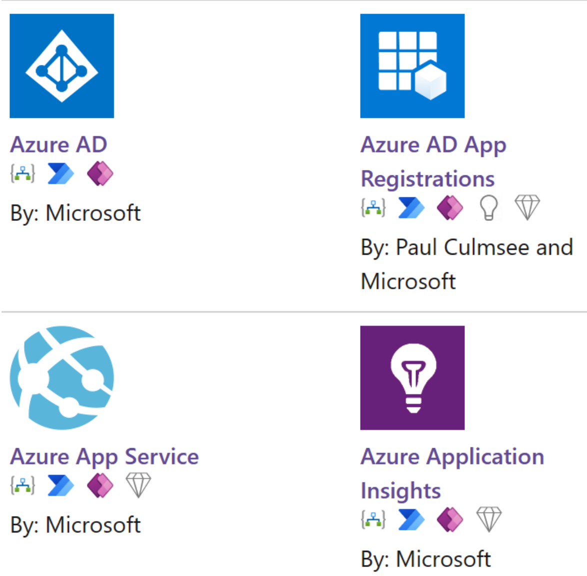
Connect your cloud flows to apps, services, and data

Standard
Connect between a wide range of cloud apps, services, and data

Premium
Expand the cloud apps, services, and data you can connect to

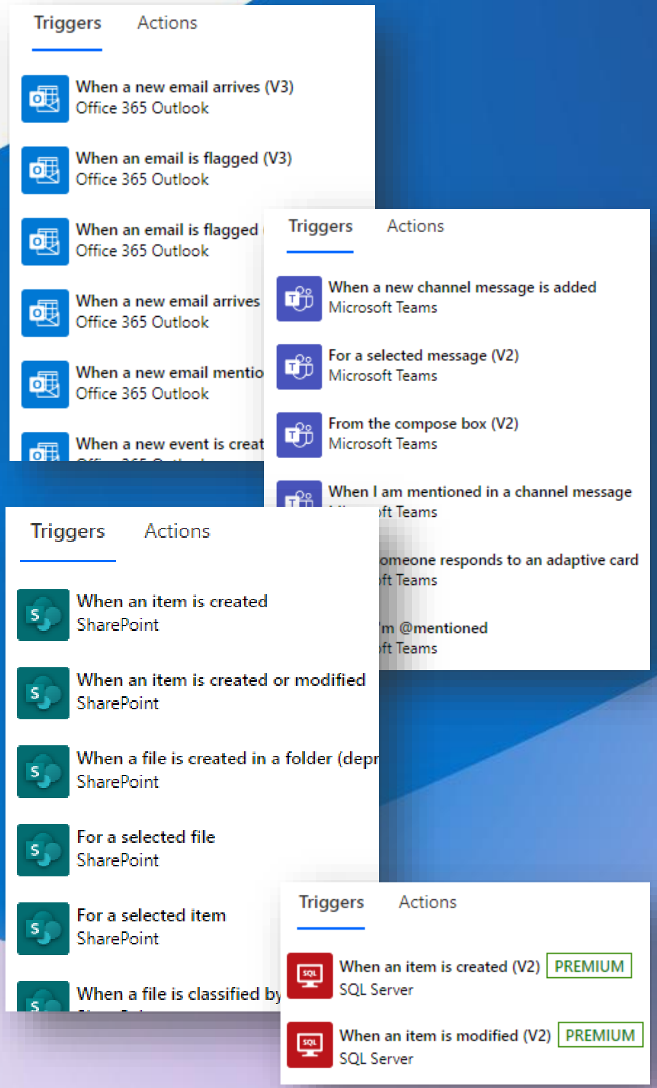
Custom
Create custom connectors from scratch or build from an OpenAI file or URL

[Connector reference overview | Microsoft Learn]



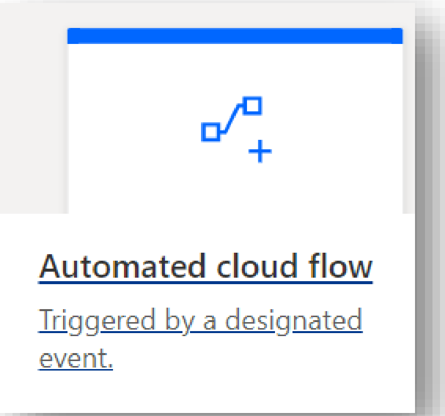
What is a trigger?

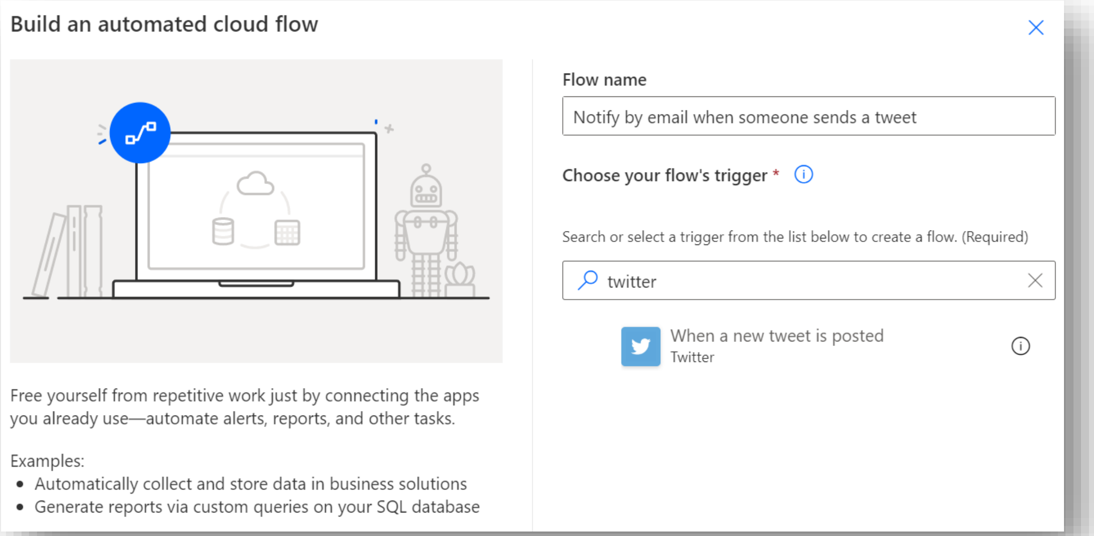
- A trigger is an event that starts a cloud flow. For example, if you want to get a notification in Microsoft Teams when someone sends you an email, in this case, receiving an email is the trigger that starts this flow.
- Triggers can be started instantly (manual), on a schedule, or automatically when an external event, such as when an email arrives
- Flow can have only one trigger



Automated cloud flows

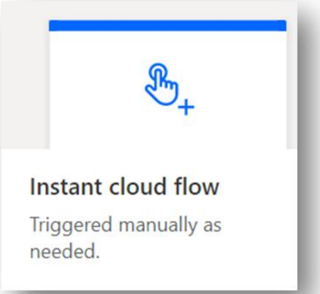
- Cloud flow that performs one or more tasks automatically after it's triggered by an event.
- For example, flow that notifies you by email when someone sends a tweet that contains a keyword you specify. In this example, sending a tweet is the event, and sending mail is the action.

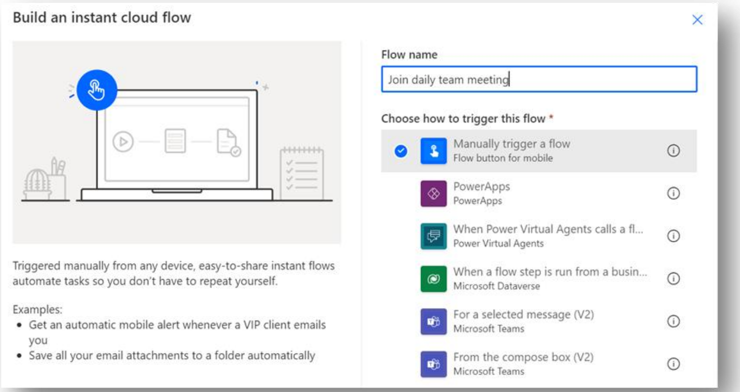


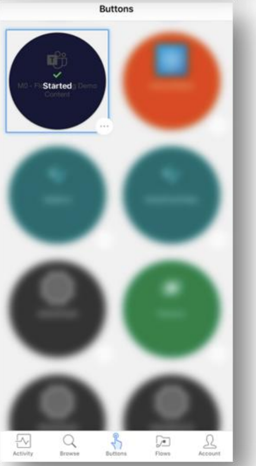


Instant (manual) cloud flows

- Cloud flow that performs one or more tasks after it's triggered manually
- For example, flow with a tap of a button on your mobile device, to remind your team to join the daily team meeting. You can trigger these flows manually from any device.
- Another instant flow example could be starting flow by clicking a button in Power Apps

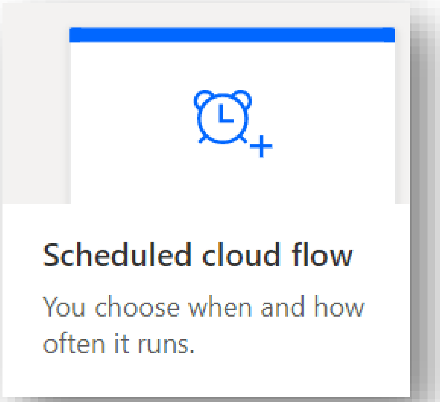


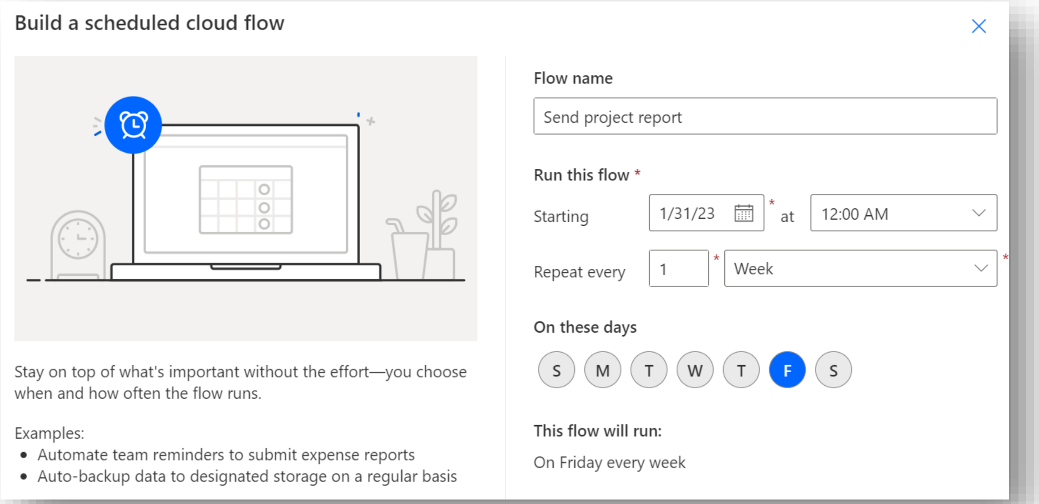




Scheduled cloud flows

- Cloud flow that performs one or more tasks after it's triggered by defined schedule like
  - Once a day, an hour, or a minute.
  - On a date that you specify.
  - After a number of days, hours, or minutes that you specify.
- For example, run a flow once per week to send a project report



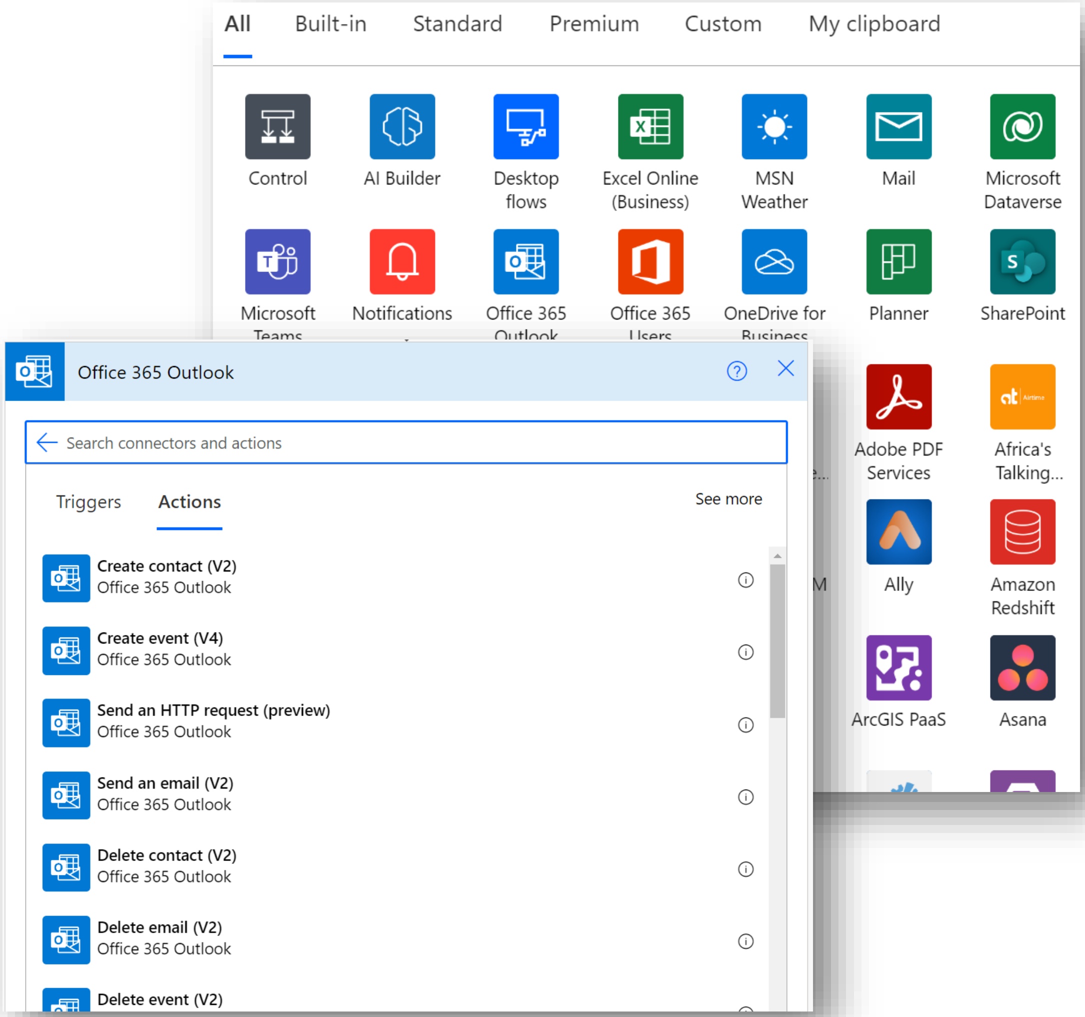


What is an action?

Flow action defines an operation flow will execute in one step of the flow, for example, post a message to Team channel

Flow can have multiple actions from any type of available connectors

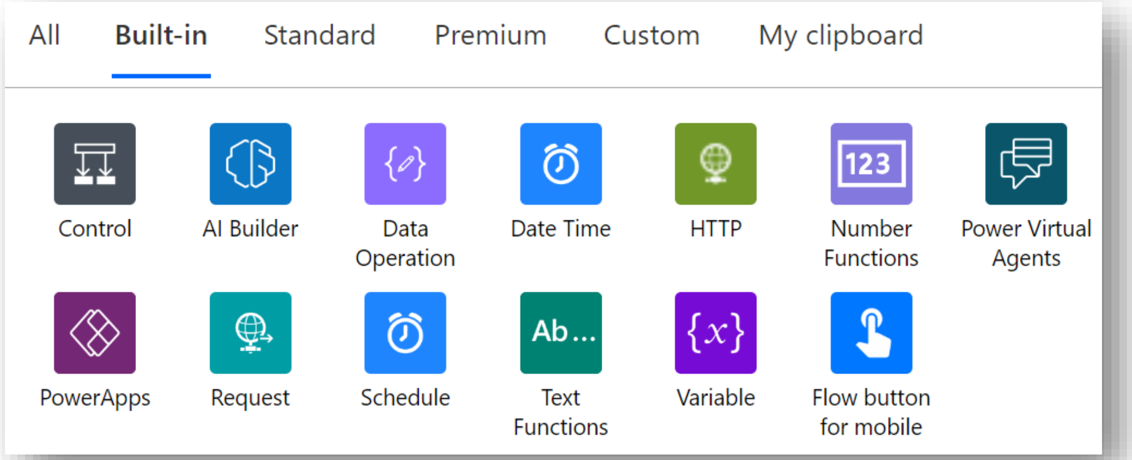
Some actions offer advanced options like sends an email message as high priority or add attachments



https://learn.microsoft.com/en-us/power-automate/multi-step-logic-flow

Built-in Actions

- You can add more control and logic to the flow by using built-in actions
- Most common built-in actions are:
  - Control like Condition and Apply to Each
  - Data operations like Filter and Parse JSON
  - Variable actions



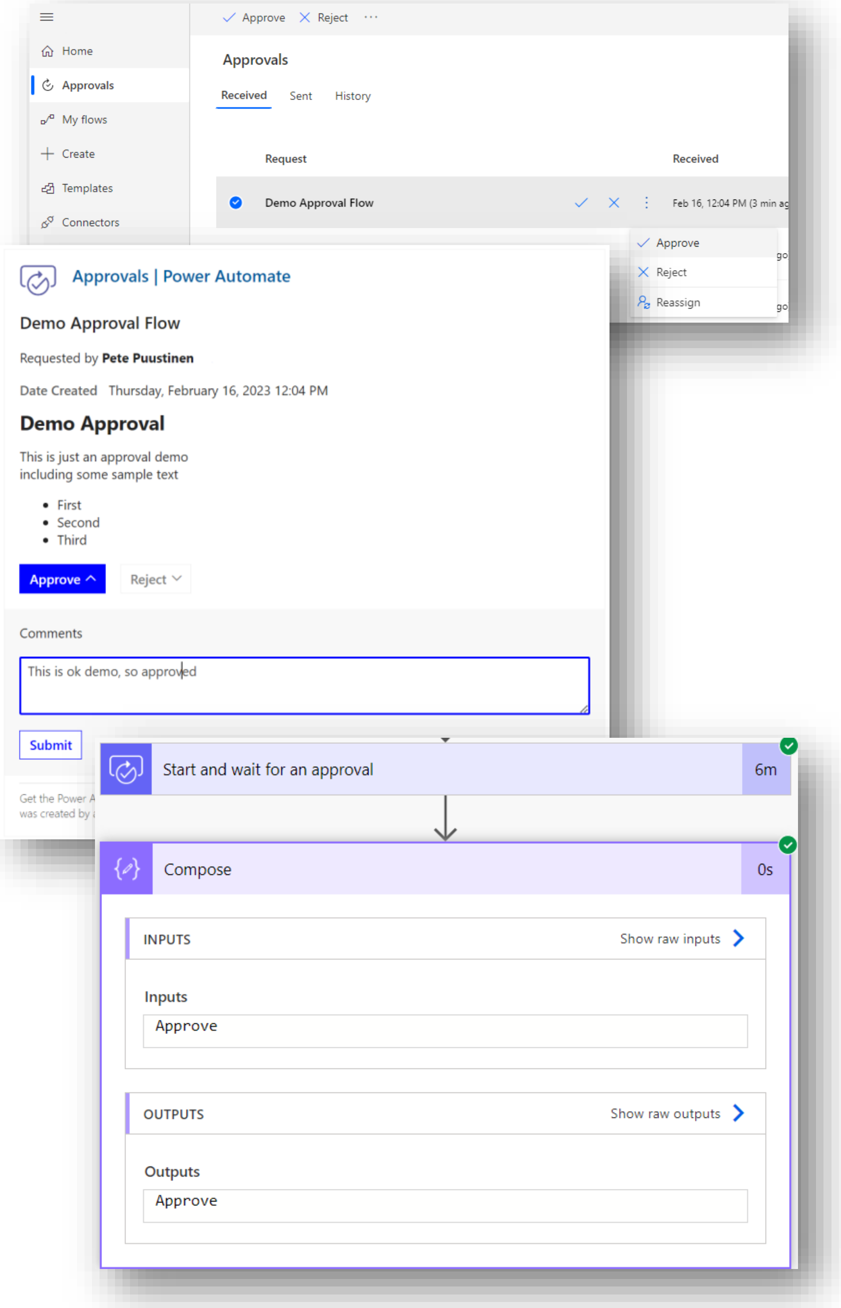
Power Automate Approvals

Whether you need written acknowledgment from your manager or a formal authorization from a diverse group of stakeholders, getting things approved is part of almost every organization.

With the approvals capability in Power Automate, you can automate sign-off requests and combine human decision-making for workflows. Some popular cases where approvals can be used include:

- Approving vacation time requests.
- Approving documents that need sign-off.
- Approving expense reports.

When you submit an approval in a flow, approvers are notified and can review and act on the request.



Advanced Actions

- Condition
- Scope
- Apply to each
- Variables
- Switch
- Do until
- Terminate
- Parallelism

https://learn.microsoft.com/en-us/power-automate/use-expressions-in-conditions

Condition

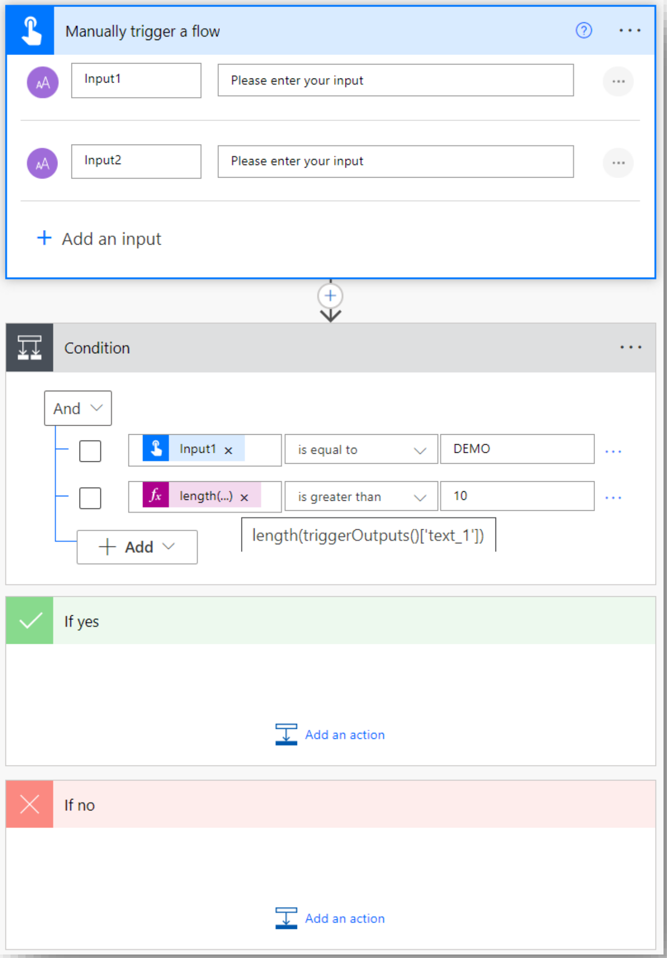
Use a condition to specify that a cloud flow performs one or more tasks only if a condition is true or false

Conditions include operators (basic mode):

- Contains / Does not contain
- Is equal to / Is not equal to
- Is greater than / Is greater than or equal to
- Is less than / Is less than or equal to

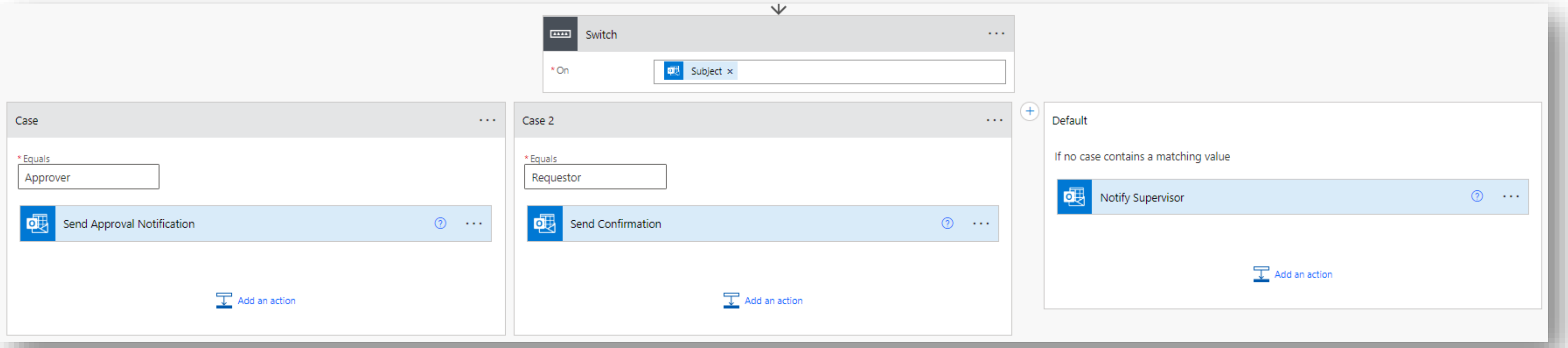
You can also use more than one expression, such as “If a parameter equals [X value or Y value]”

It is also possible to specify an expression in your conditions. This allows you to add more logic and control for conditions



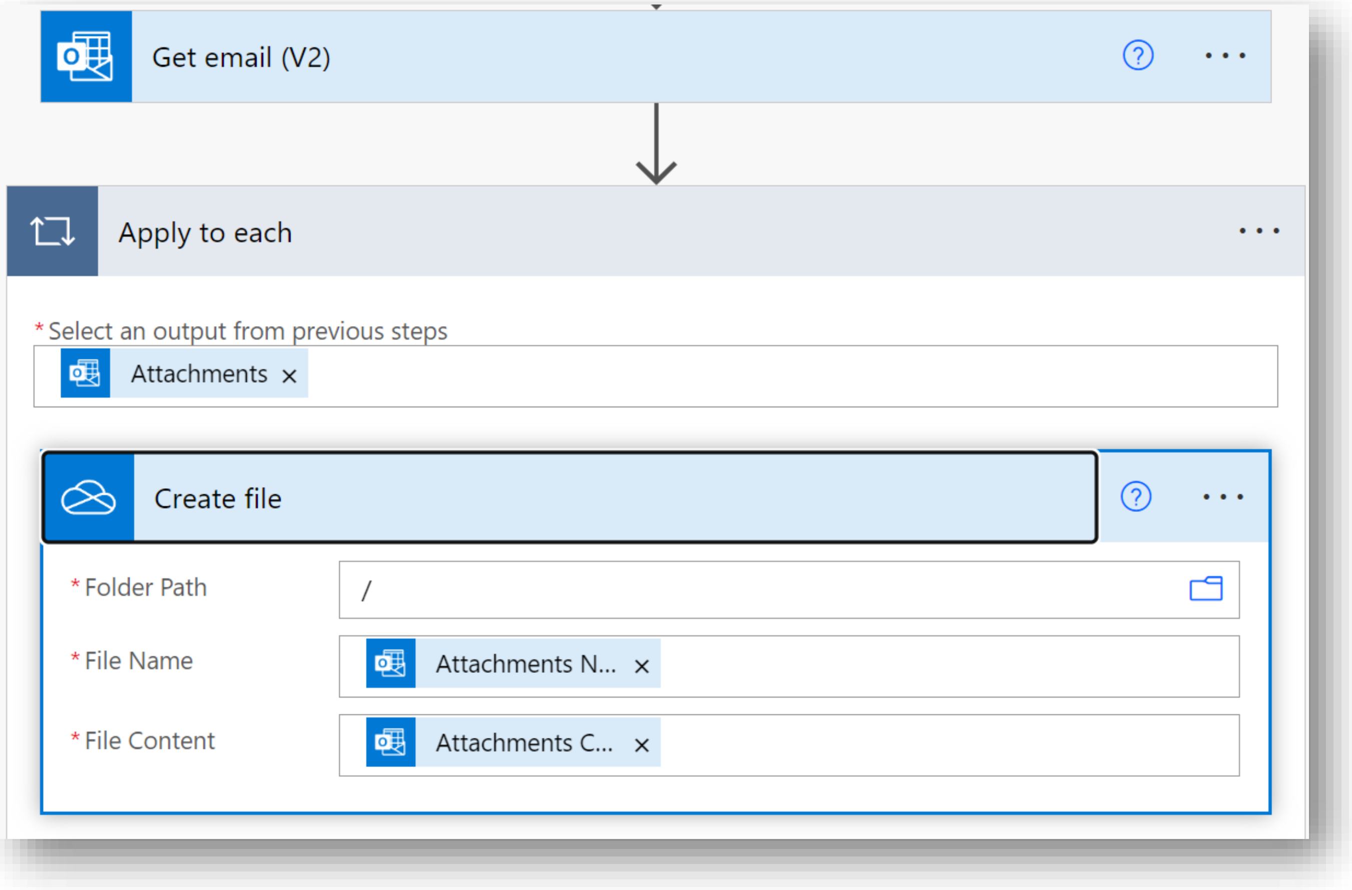
Switch

Use switch case built-in action to have multiple conditions



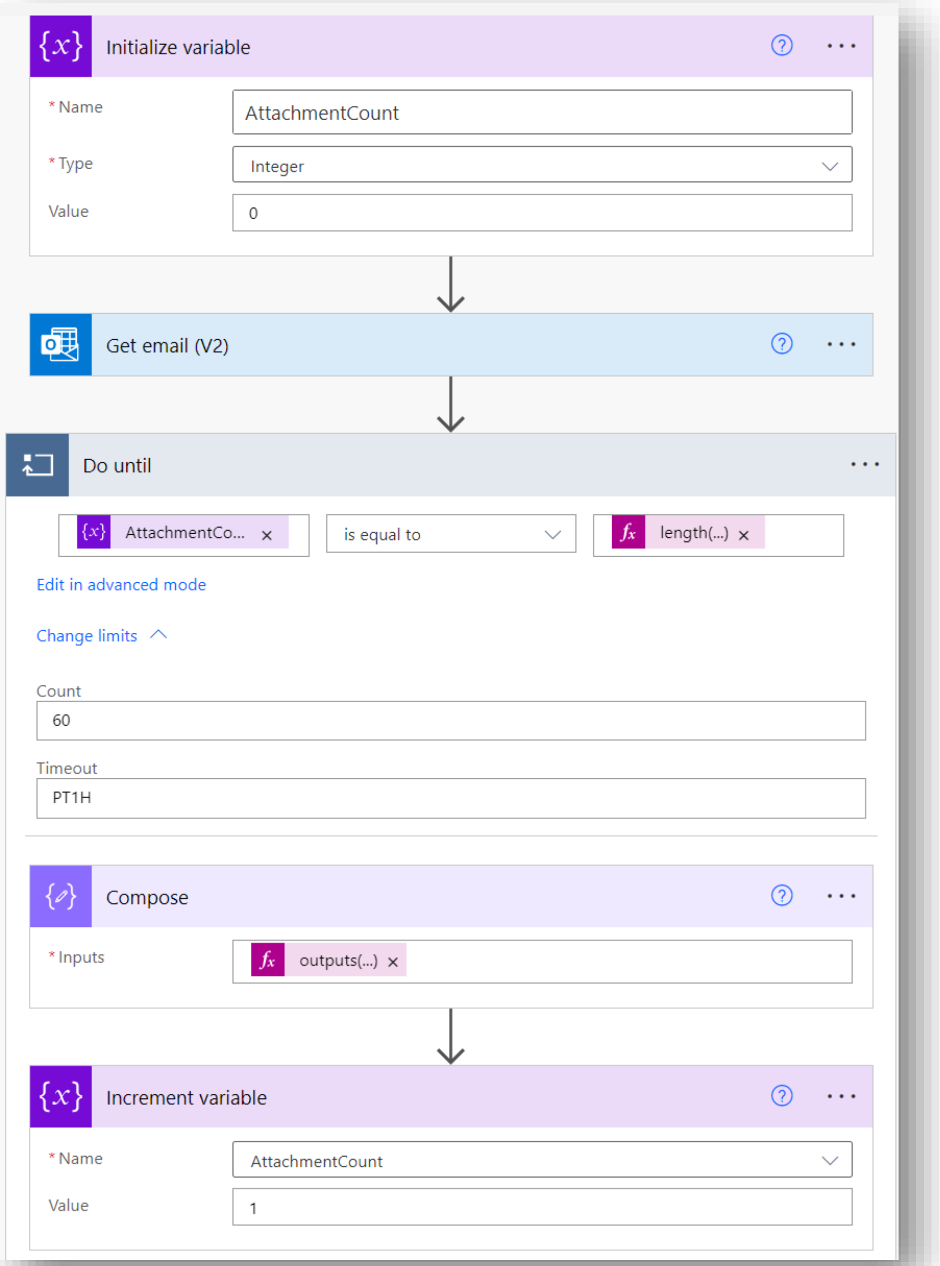
Apply to Each

Use Apply to each loop to process a list of items. For example, you could use apply to each to save email attachments to OneDrive



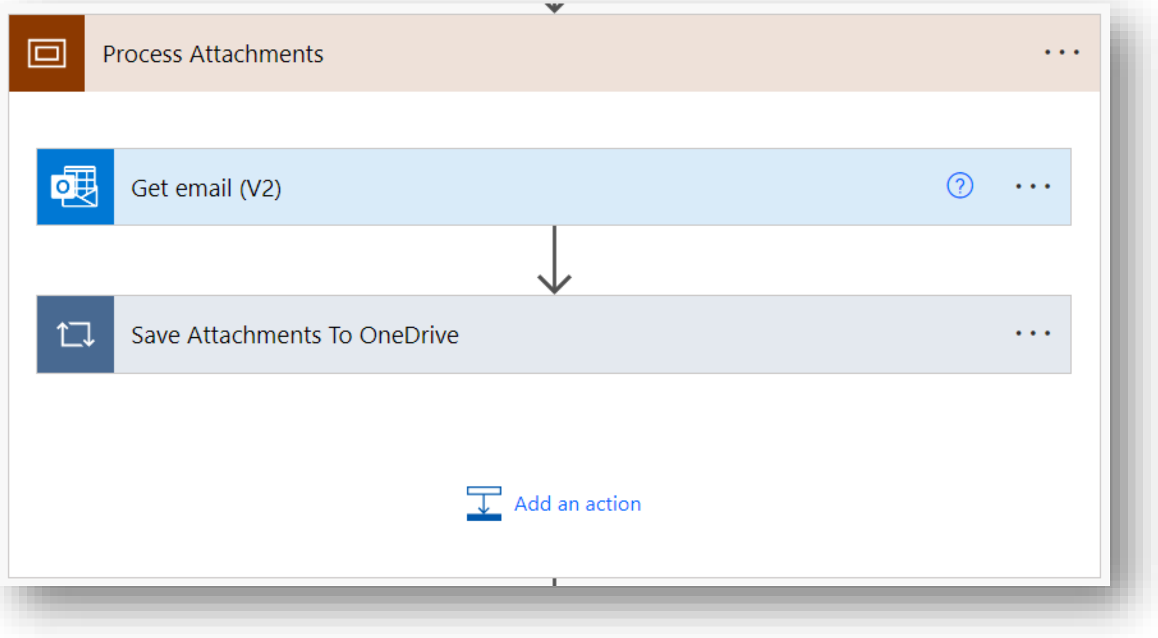
Do Until

- Use Do Until loop to run actions in the loop as many times as the defined condition
- If the condition is met, then loop exits i.e. it will run in the loop as long as condition is not true
- You can change the maximum count (5000) and timeout limits



Scope

- Use scopes to group actions and keep you flow better organized
- Also easier to move multiple actions around and even copy-paste
- Used also in the error handling (discussed in more detail later)



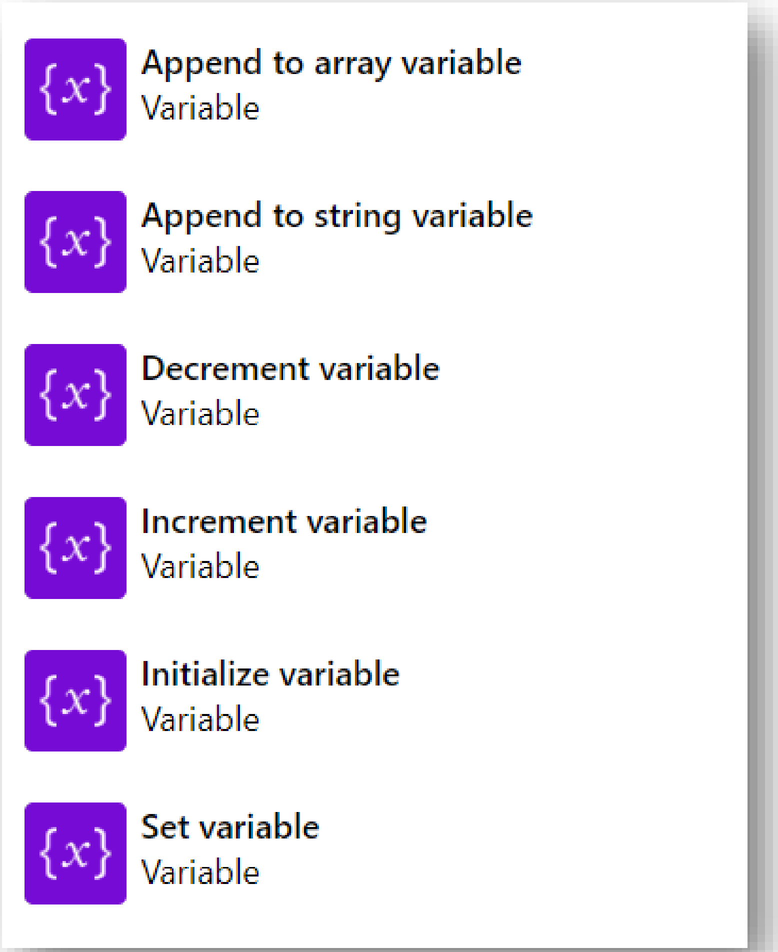
Variables

Variables can be used to store values in your cloud flows.

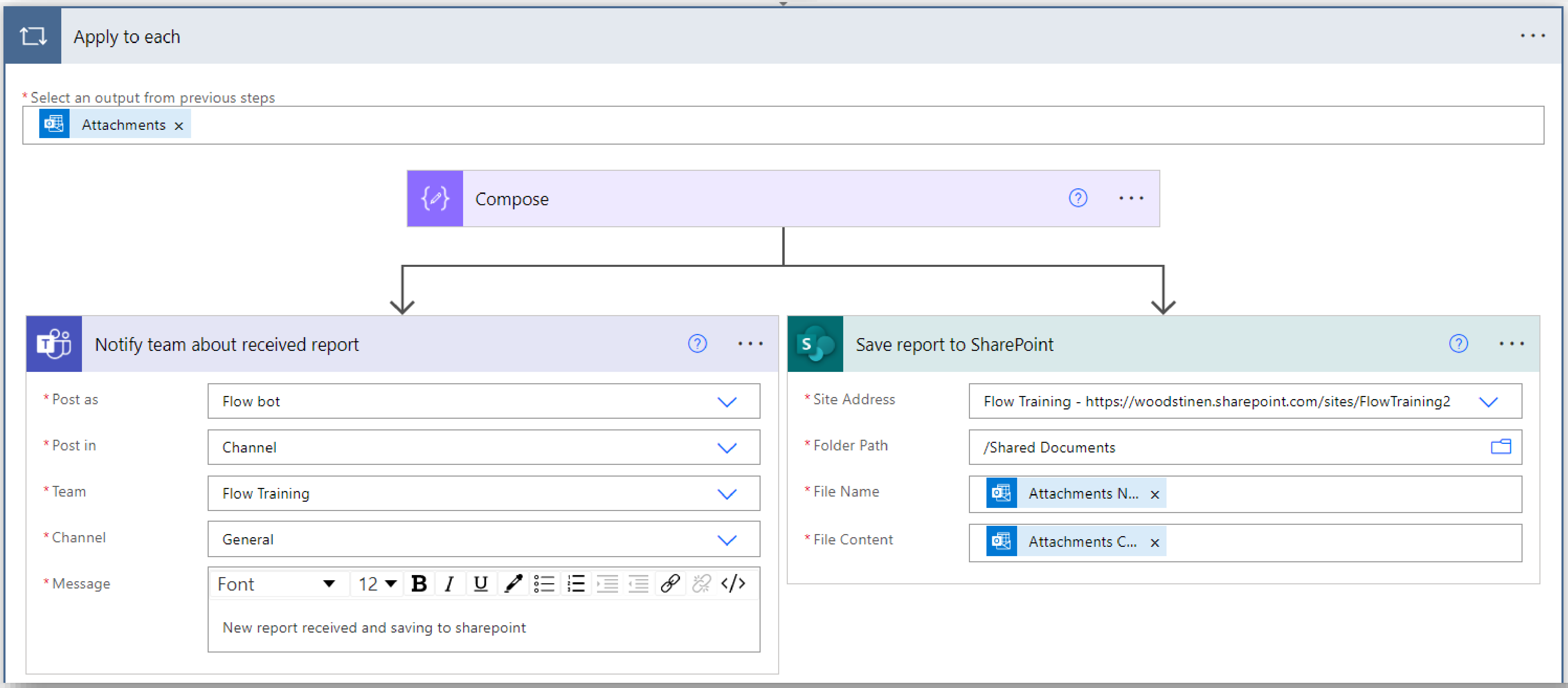
For example, track the number of times a loop runs. To iterate over an array or check an array for a specific item, you can use a variable to reference the index number 'apply to each' array item.

Supported datatypes are:

- Boolean
- Integer
- Float
- String
- Object
- Array



Run in Parallel - two or more steps that run at the same time



Connectors

Popular Standard connectors

Excel Online (OneDrive)	Office 365 Outlook	OneDrive for Business	Microsoft Teams	Microsoft To-Do (Business)
Office 365 Users	Microsoft Forms	Power Apps Notification V2	SharePoint	Planner

[List of supported connectors | Microsoft Power Automate]

Popular Premium connectors

Dataverse

Adobe PDF Services

SQL Server

DocuSign

Azure OpenAI

Calendy Forms

ClickSend SMS

ServiceNow

[List of supported connectors | Microsoft Power Automate]

Independent Publishers

- Connectors that are developed, tested, published, and supported by connector developers that are independent from the end service or API.
- Independent Publishers include Microsoft employees, MVPs, developers, and other community members.
- Independent publishers have to go through a certification process for the connectors they wish to publish.

They can be free but may be blocked by your organization.

OpenQR

IQAir

[List of supported connectors | Microsoft Power Automate]

Your First Flows

How to start using Power Automate

	Start with...	Progress to...	Automation Team...
Target users	Personal	Team	Company
Complexity	Low	Medium	High
What to automate?	Partial Workflow	Single Workflow	Multiple Workflows

Example: Translating processes to Cloud Flows

Invoice Document Process

A building management uses vendors to perform services at their buildings, such as cleaning properties after tenants leave and providing landscaping services at their properties. After the jobs are completed, vendors send invoices to the building management company. The building management company might want to automate processing invoices from vendors.

Currently the process is as follows:

1. Invoices are received as email attachments. The management company uses Microsoft Exchange for processing emails.
2. These attachments are downloaded and stored in Microsoft SharePoint.
3. Each invoice is sent to a specific person for approval. Approval is made in Microsoft Teams.
4. Once approved, the invoices are entered into the company’s ERP system. The management company uses Oracle. Once entered, a confirmation email with the invoice number must be sent back to the vendor.

Example: Translating processes to Cloud Flows

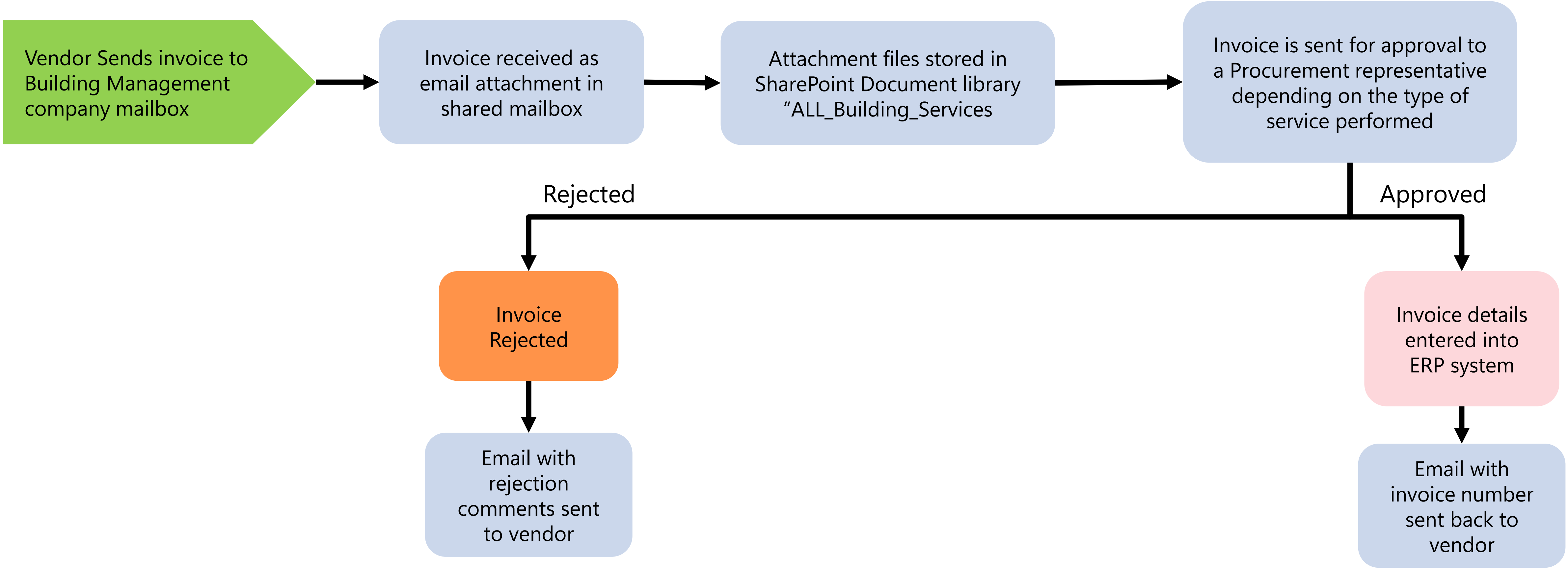
Invoice Document Process

Automating a process like this requires interaction with multiple services. There are likely different accounts being used with each service, and different data operation that are being executed. In this example, we would be working with at least four different services.

- **Office 365 Outlook:** First, you would need to monitor a specific Microsoft Exchange mailbox.
- **Microsoft SharePoint:** Attachments are downloaded and saved to a specific folder in Microsoft SharePoint.
- **Microsoft Teams:** Invoice approval requests are sent to managers in Microsoft Teams, where they can either approve or deny the request.
- **Oracle:** Once approved, the new invoice is created in Oracle. Details of the invoice are stored so they can be used later.
- **Office 365 Outlook:** A confirmation email is sent using a dedicated mailbox to the company that submitted the invoice.

Example: Translating processes to Cloud Flows

Invoice Document Process Mapping



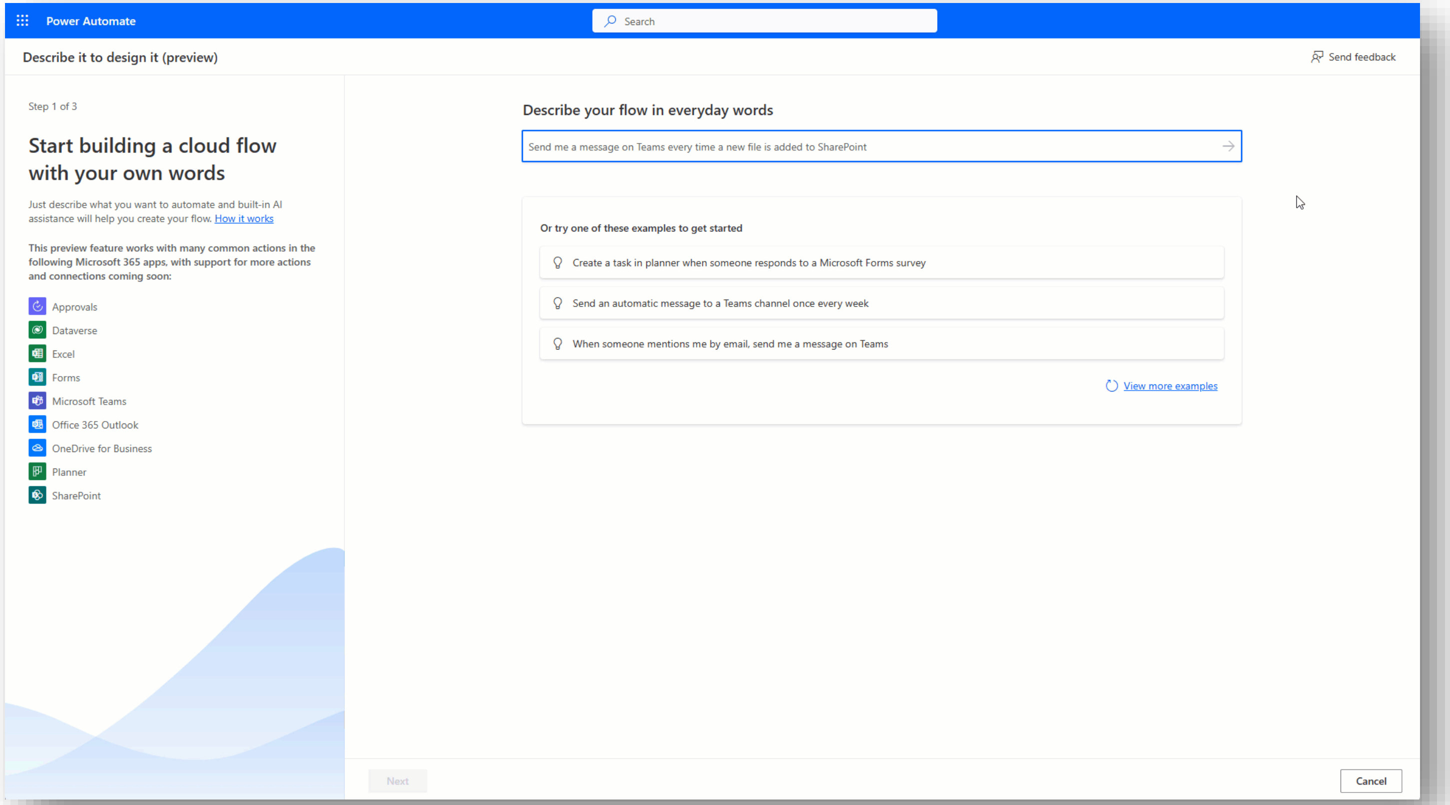
Copilot in power automate

Transform natural language to Automation

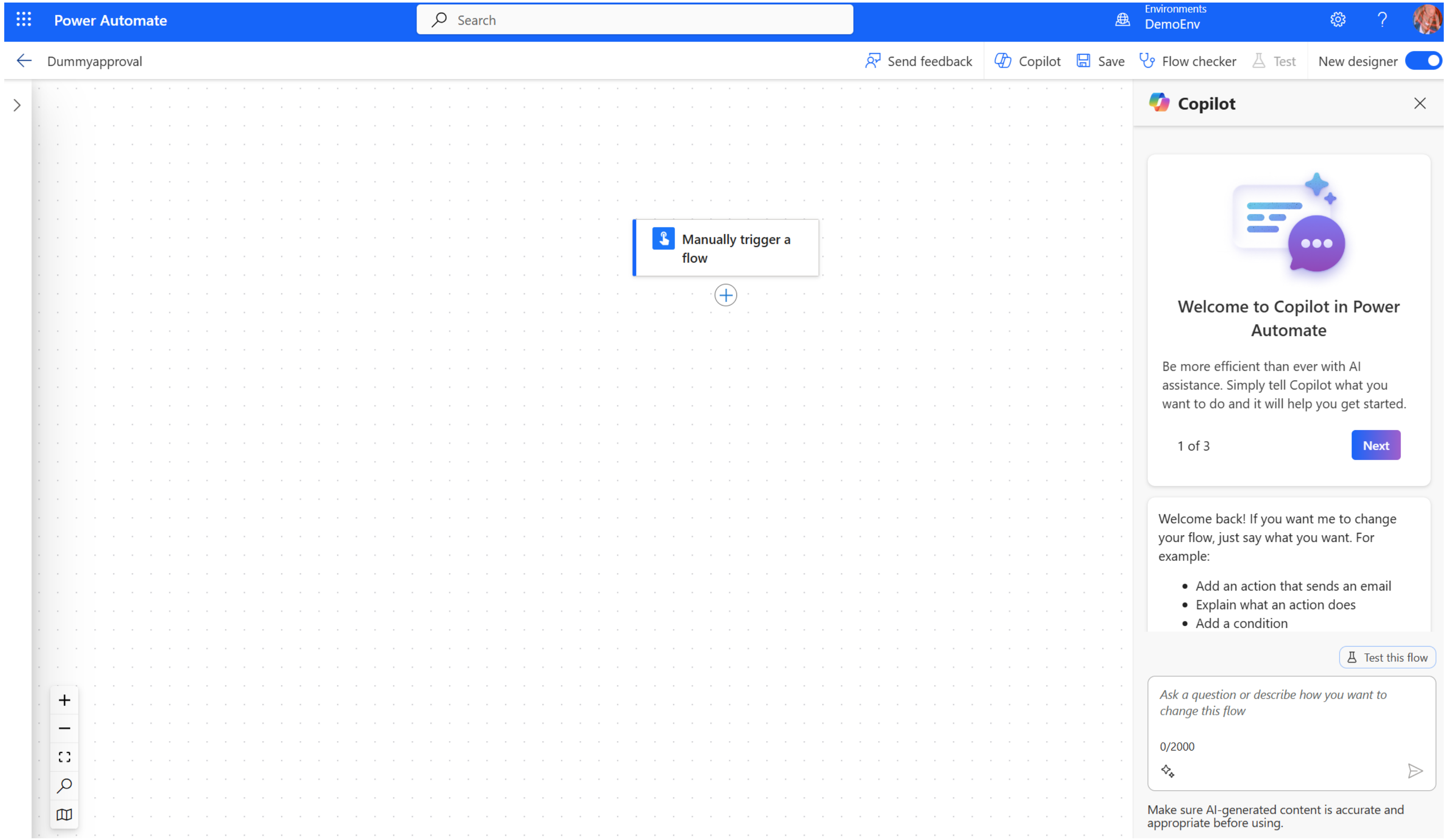
Bring ideas to life with AI powered automation development

Now you can describe what you want to automate in a sentence, and an AI-based copilot will build your flow in seconds

This new way to create flows runs on OpenAI Codex, an AI model descendant of GPT-3 that can translate natural language to code, in this case, Power Automate cloud flows



Create, edit and explain with copilot



Now you can **describe what you want** to automate in a sentence, and an AI-based copilot will build your flow in seconds

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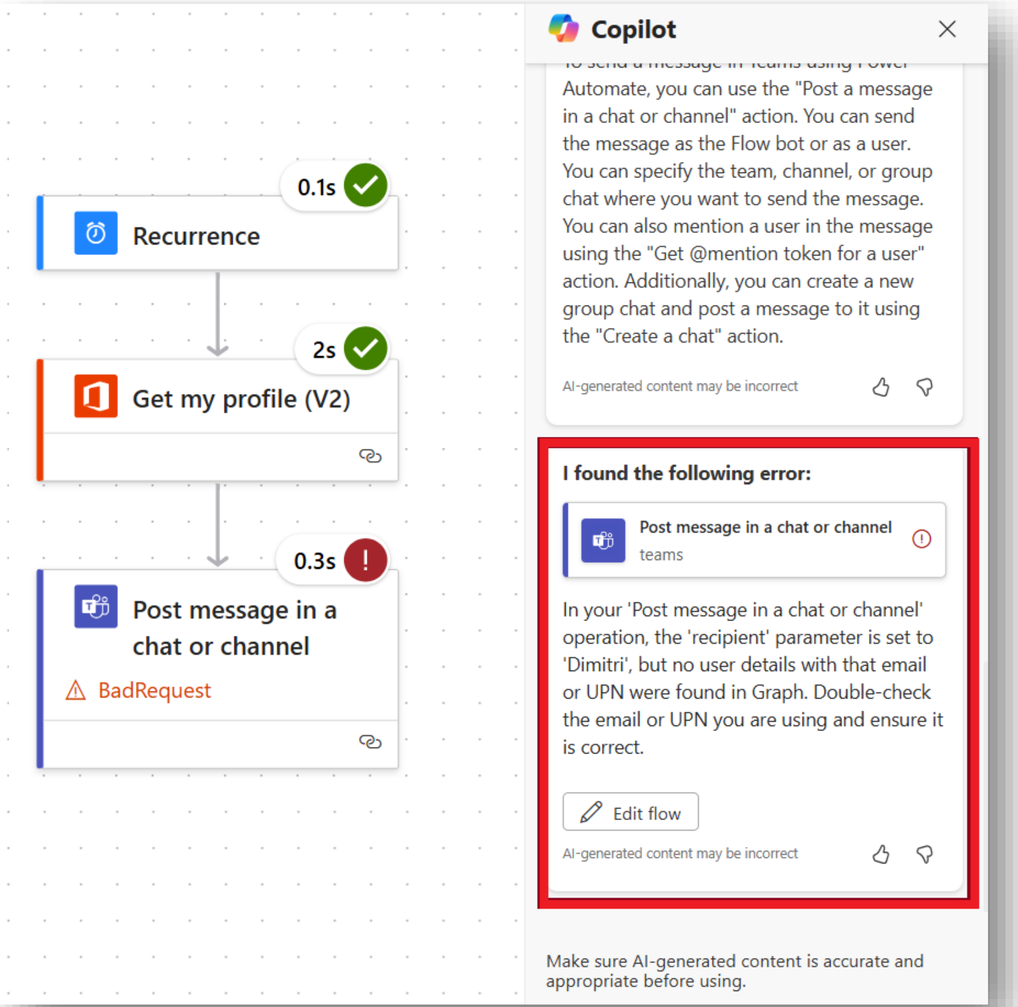
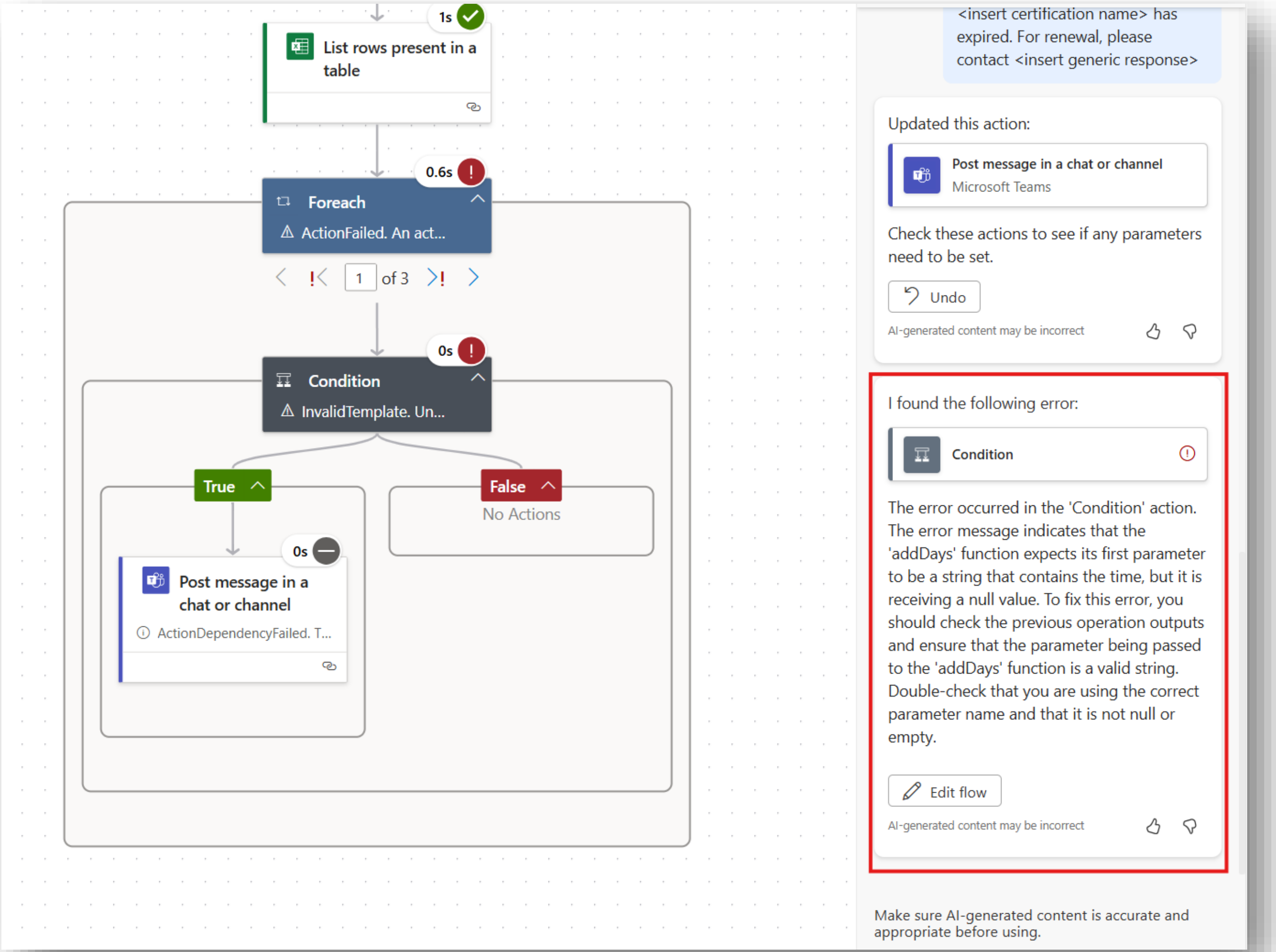
Troubleshoot with copilot

Identify and resolve errors that might occur during the testing of cloud flows





Microsoft Power Platform

Thank you

Sing Kwang Eng
v-kwasing@microsoft.com

Please help to fill up feedback survey with below URL or QR code above

https://forms.office.com/r/NVxHNTNSXi